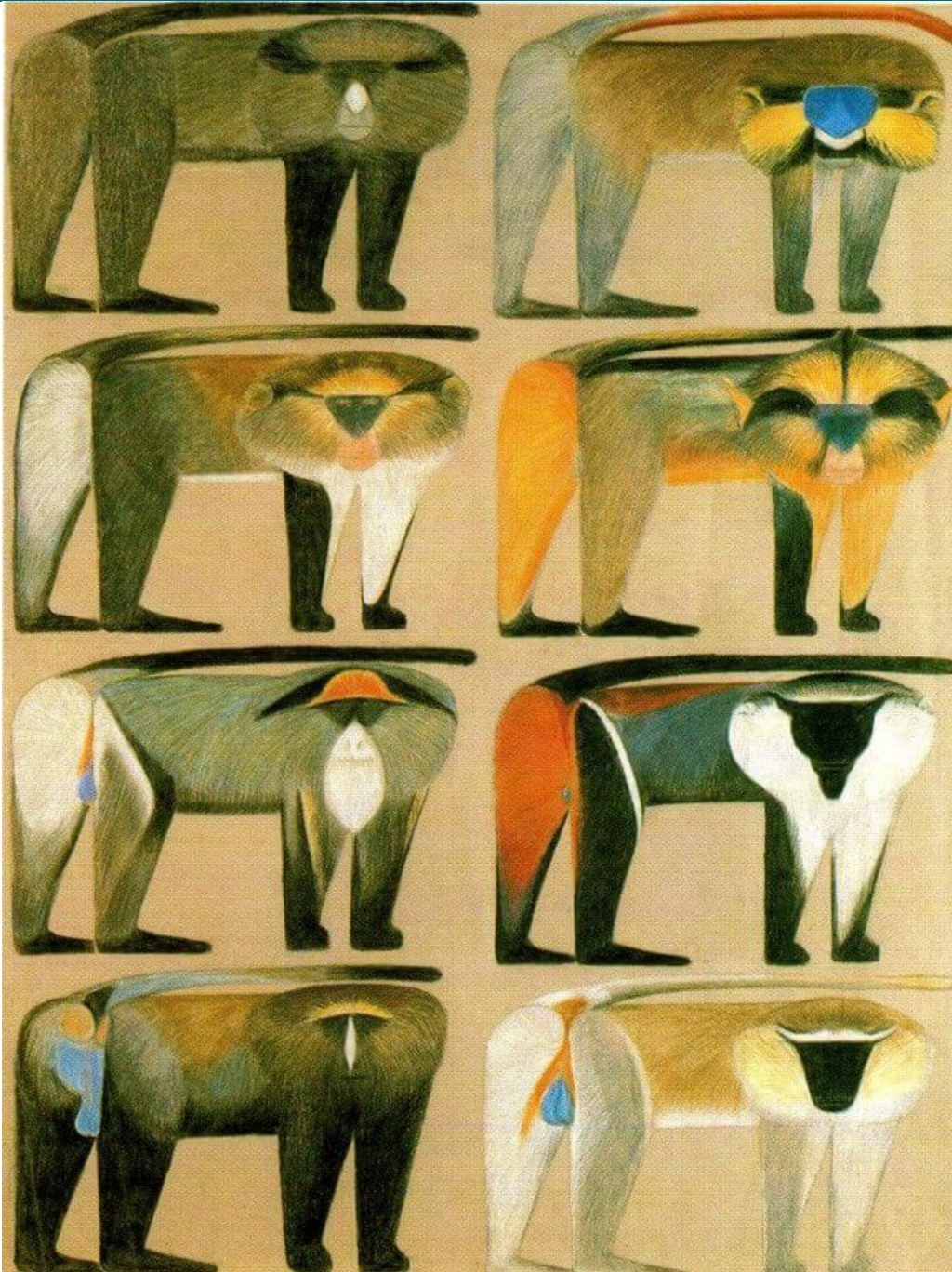




How locally adapted are modern humans?

Dr. Laura van Holstein

lavanho@emory.edu

Room 215

Coldes + place on earth
where humans continuously
live (-57°F)

Oymyakon (Russia)



Mbendjele BaYaka (CAR & RC)

High altitudes

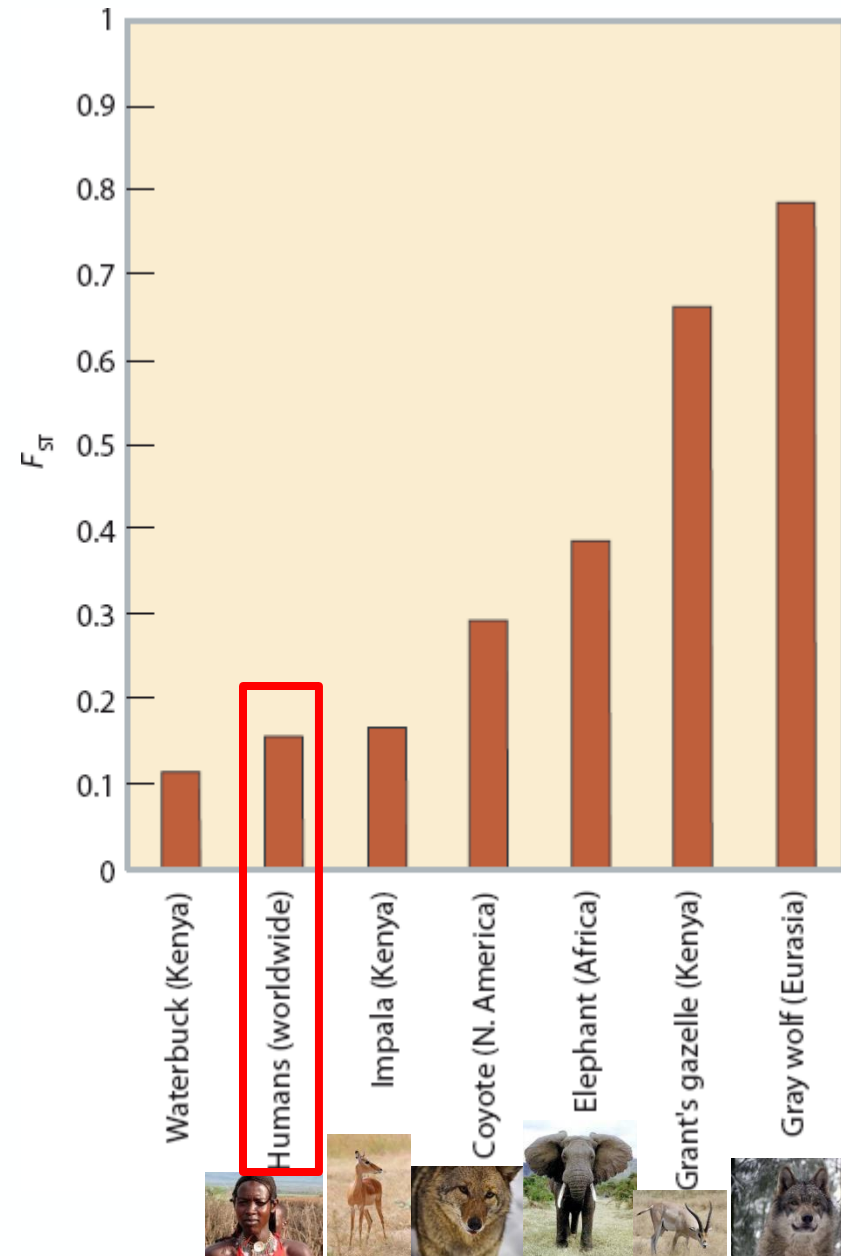
Quechua (Peru)





Bedouin (North Africa)

And yet...

[Review: Lecture 7](#)



Human
genetic
diversity
related
to cows!

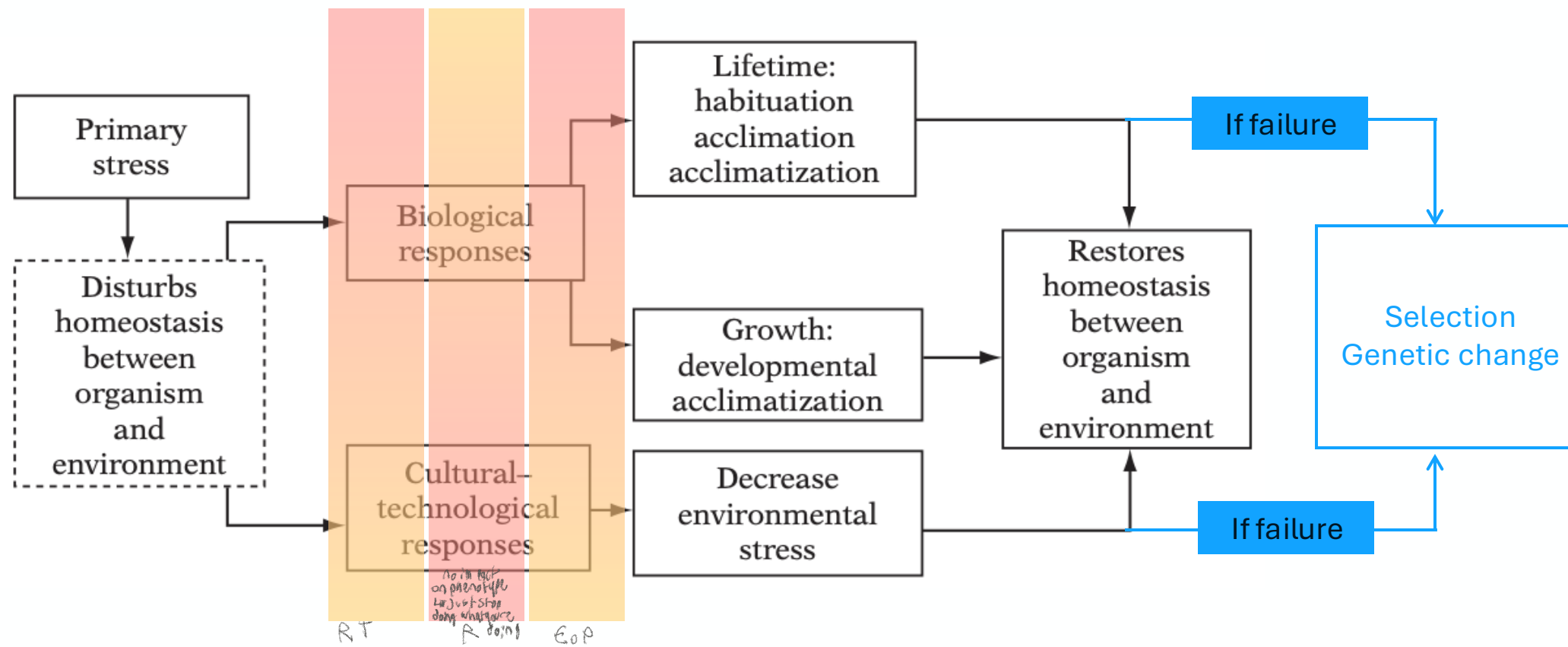
The key question: in such a genetically homogenous but globally distributed species, does **local adaptation** effectively only happen through **short-term processes** (culture, homeostatic responses, and acclimatization)?

Two main takeaways

1. The human genome is (uniquely?) **buffered** from intense selection. Most human genetic variation is selectively neutral. Culture plays an important role, but there are also arguments that humans are exceptionally developmentally adaptable.
2. However... there are stressors that surpass short-term processes! One of the most important stressors (measured in the number of known genetic adaptations *and* the strength of selection) is **disease**.

* considering low genetic diversity

Buffering



Response time:



Quicker Slower

Reversibility:



Low High

Effect on phenotype:



Weak Strong

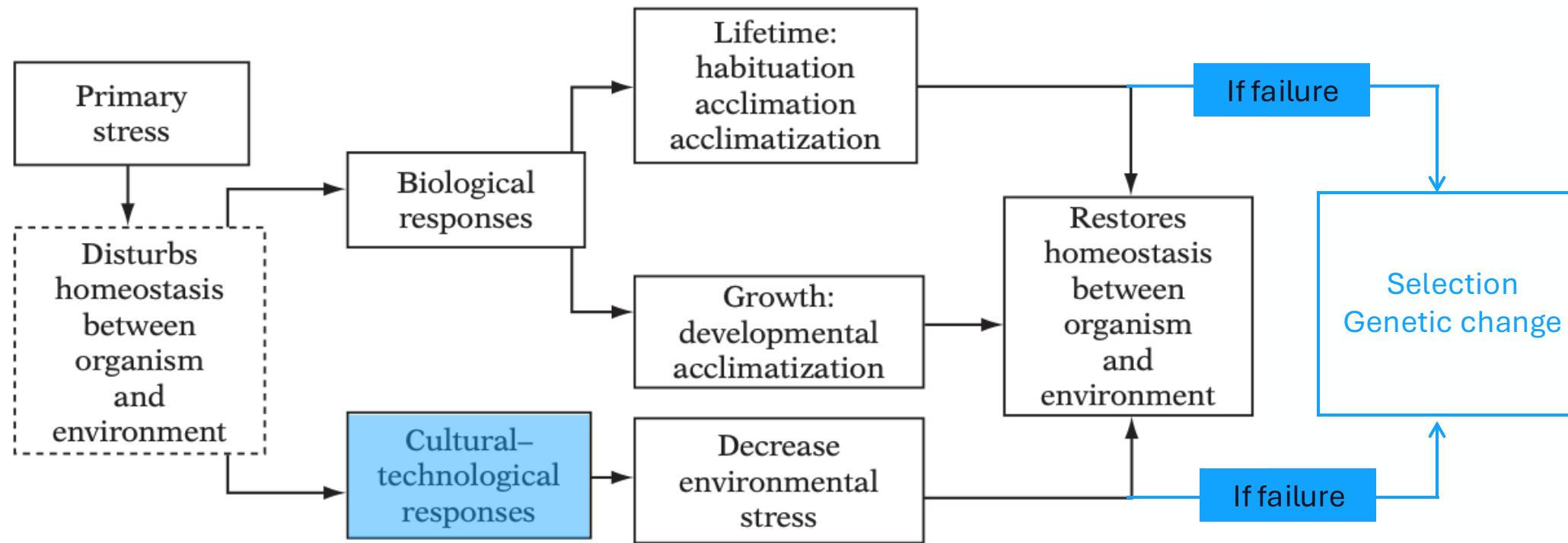
Efficacy:



Weak Strong

?

Buffering



Efficacy:



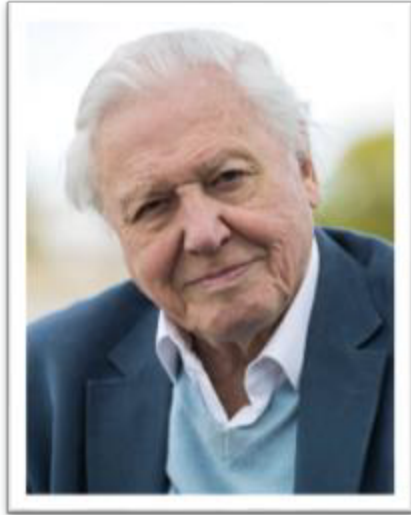
Weak

?

Strong

It's concepts about

Buffering



“We are the only species that have **put a halt to natural selection**, of its own free will...We stopped natural selection as soon as we started being able to rear 95- 99% of our babies that are born”
(Sir David Attenborough, Radio Times, 2013)

This idea wld logic
is very pervasive...

“The epoch of Darwinian evolution...ended about ten thousand years ago, when a single species, *Homo sapiens*, began to dominate and reorganize the biosphere. Since that time, **cultural evolution has replaced biological evolution** as the main driving force of change”
(Freeman Dyson, 2007)



My concept is about

Buffering



“The high level of plasticity relative to other apes allows humans to **opt out of genetic adaptation**.”

We are an animal that above all refuses to commit to specific long-term strategies. ... Human biology therefore displays an extraordinary capacity for **behavior and technology to homogenize diverse habitats and construct a narrower range of ecological niches**, further accommodated by metabolic plasticity, thus buffering selective pressures on the human genome.”

Key features of cultural responses:

- **Speed** - cultural adaptations are fast
- **Creation of microclimates**

↳ Ind. clothing

developmental plasticity of Homo erectus is debatable due to lack of data

This is true even for erectus!

Cultural buffering: microclimates



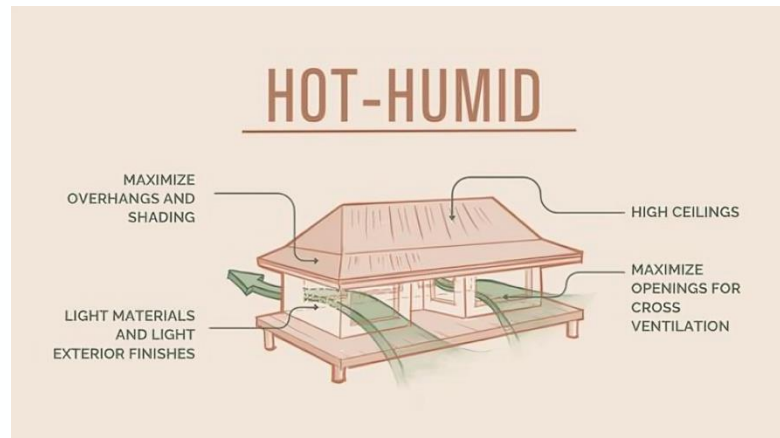
Cultural buffering: microclimates



Human structures in cold environments are very effective



Cultural buffering: microclimates



Cultural buffering: microclimates



?!

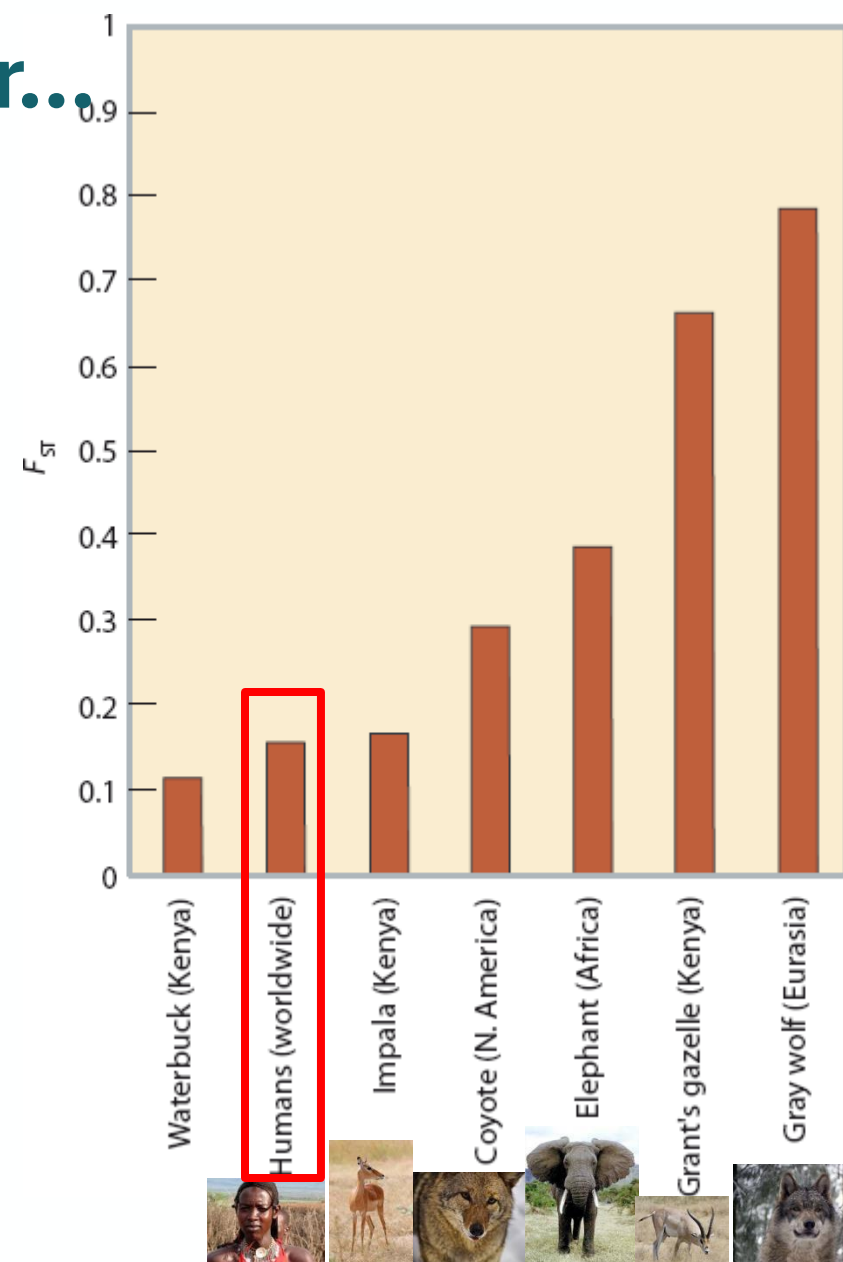


Cultural buffering: microclimates



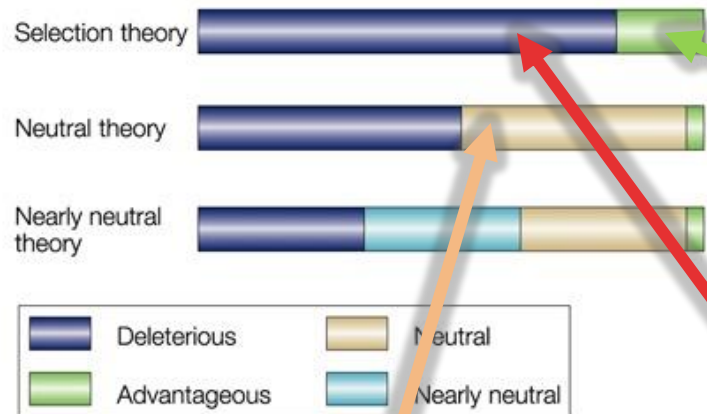
Populations *do* differ...

Review: Lecture 7



...might it all be *neutral*?

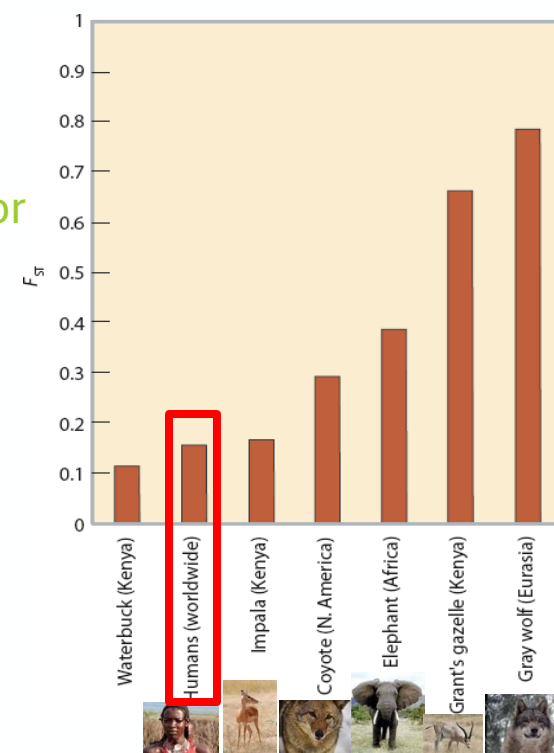
Classification of new mutants:



Selected for

Selected against

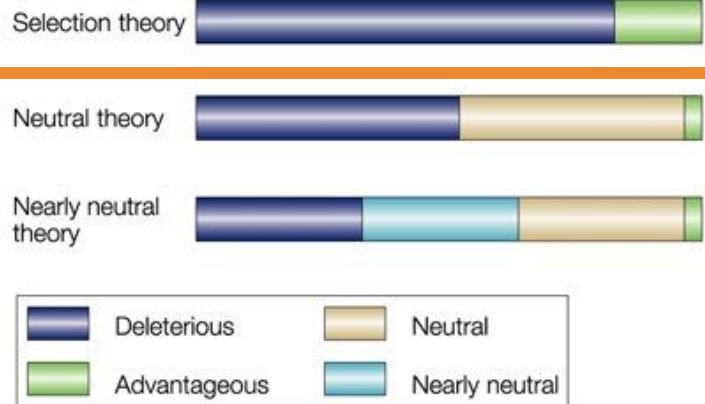
?



Motoo Kimura (1924 - 1994)

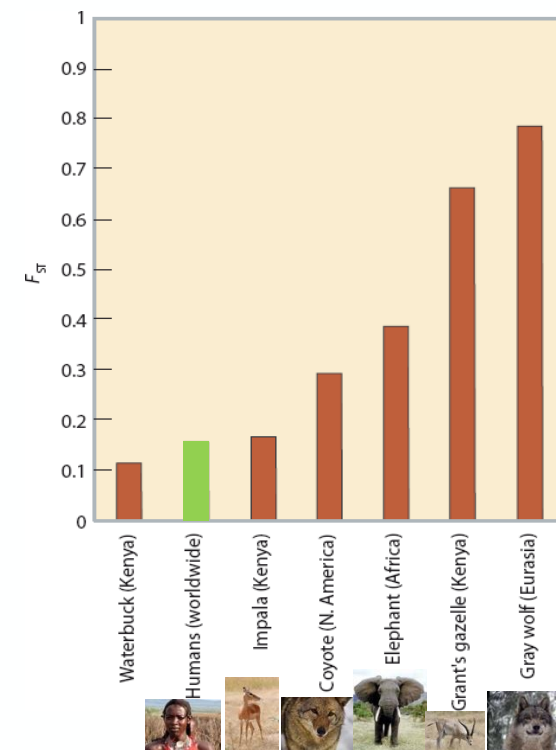
...might it all be *neutral*?

Classification of new mutants:



Selectionist/neo-Darwinian:

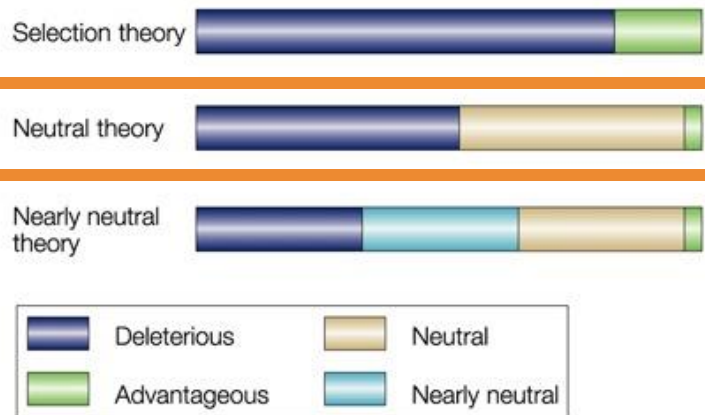
All differences between populations adaptive; due to positive selection for **advantageous** alleles and negative selection against deleterious alleles



Motoo Kimura (1924 - 1994)

...might it all be *neutral*?

Classification of new mutants:

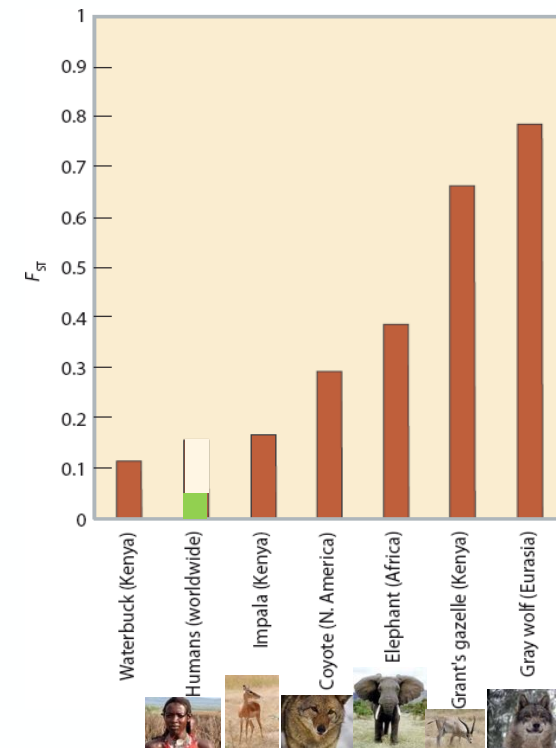


Selectionist/neo-Darwinian:

All differences between populations adaptive; due to positive selection for **advantageous** alleles and negative selection against deleterious alleles

Neutral:

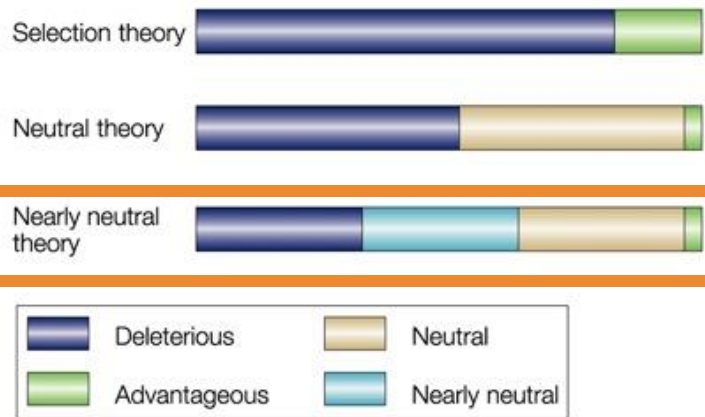
Most differences between populations **neutral**; some adaptive



Motoo Kimura (1924 - 1994)

...might it all be *neutral*?

Classification of new mutants:



Selectionist/neo-Darwinian:

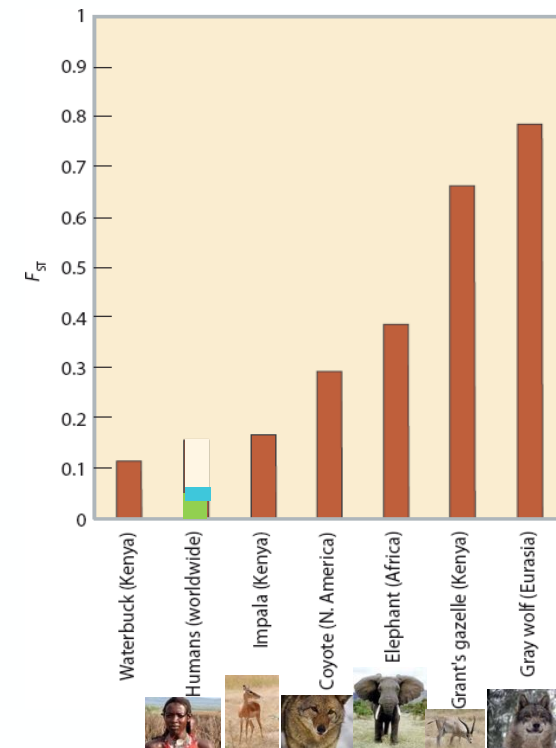
All differences between populations adaptive; due to positive selection for **advantageous** alleles and negative selection against deleterious alleles

Neutral:

Most differences between populations **neutral**; some adaptive

Nearly neutral:

Most differences between populations **neutral**; some '**nearly**' **neutral**, some adaptive



Motoo Kimura (1924 - 1994)

...might it all be *neutral*?

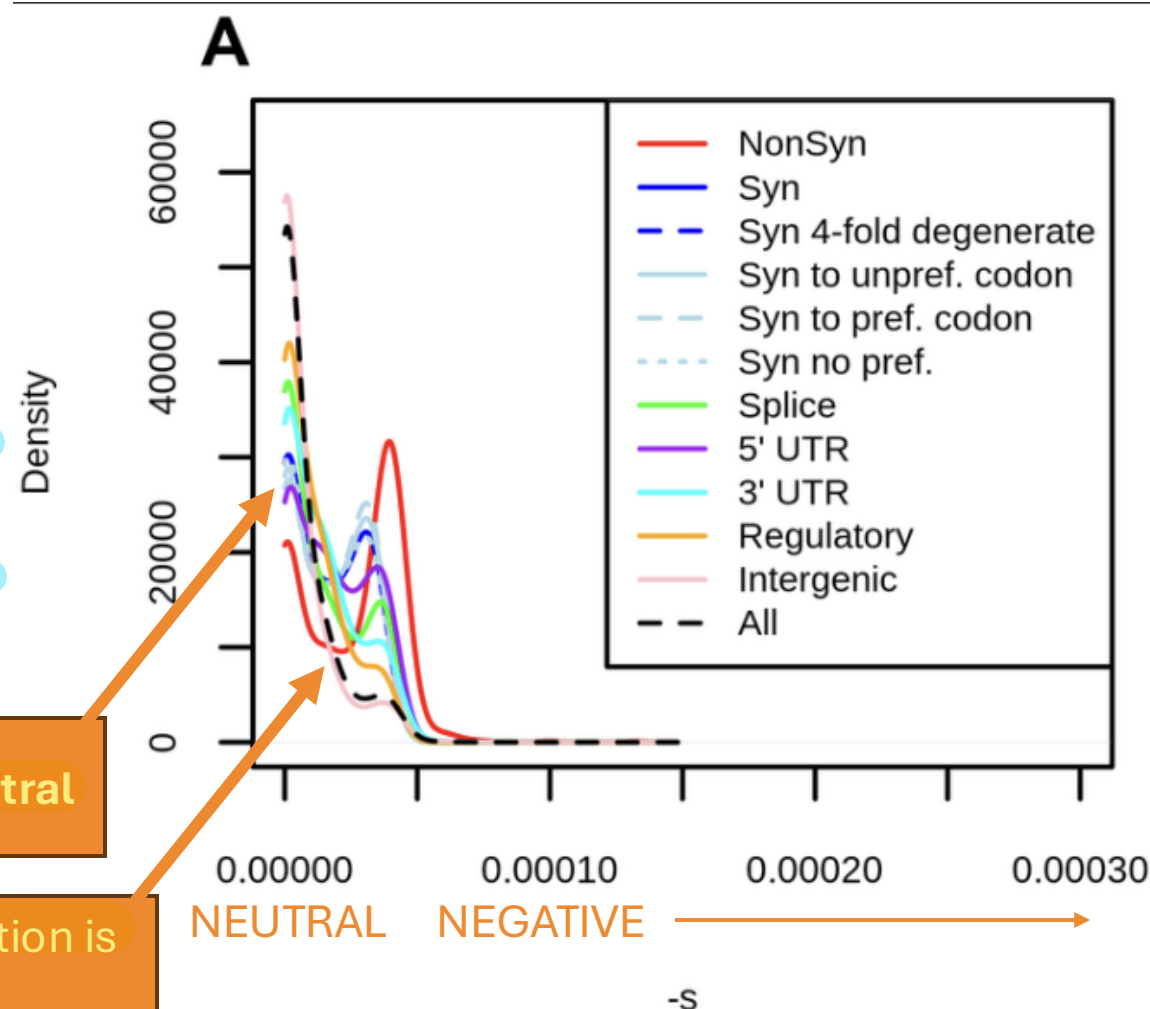
Nearly neutral theory predicts that most novel variants are approximately neutral.

What does the **human data** say?

- Segregating SNPs (i.e., >1 allele) in the Yoruba population
- Published annotations (including conservation scores, protein deleteriousness estimates and regulatory data) to score all mutations in the human genome based on how likely they are to be affected by negative selection

Vast majority of variation is **neutral**

Some nonsynonymous variation is **weakly negative**



...might it all be *neutral*?

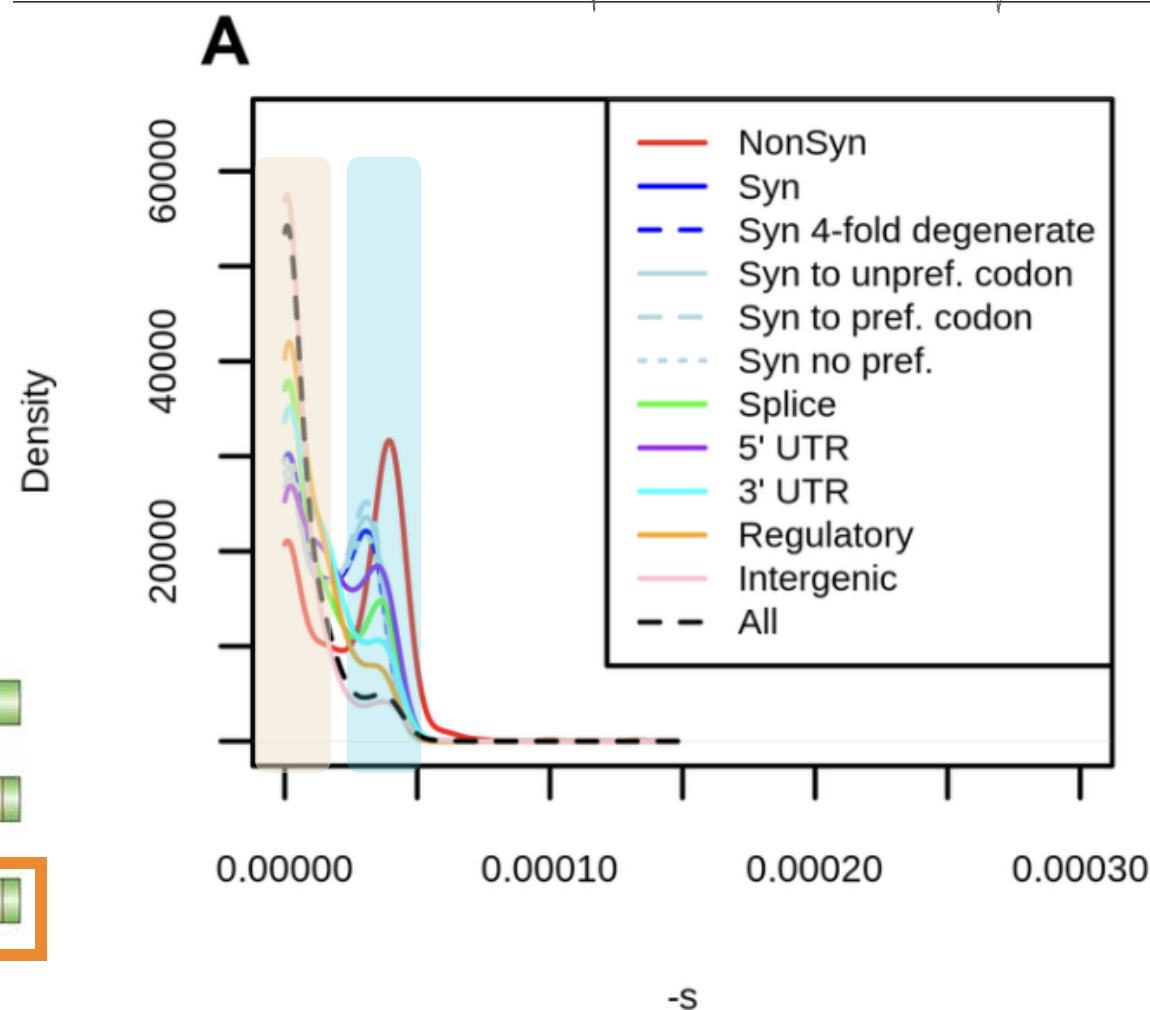
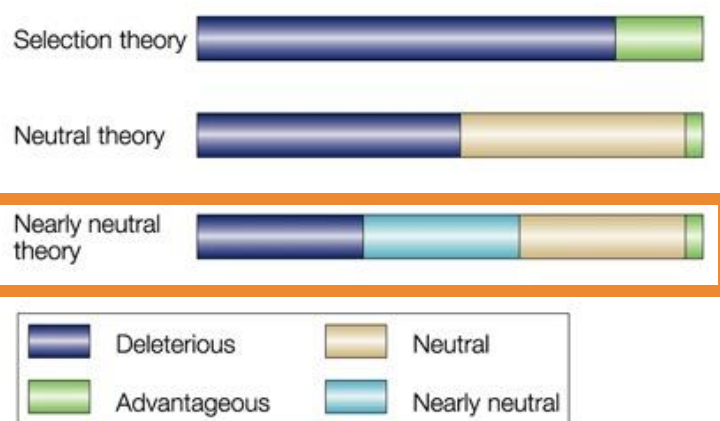
Evidence for nearly neutral theory

Why no positive effects?

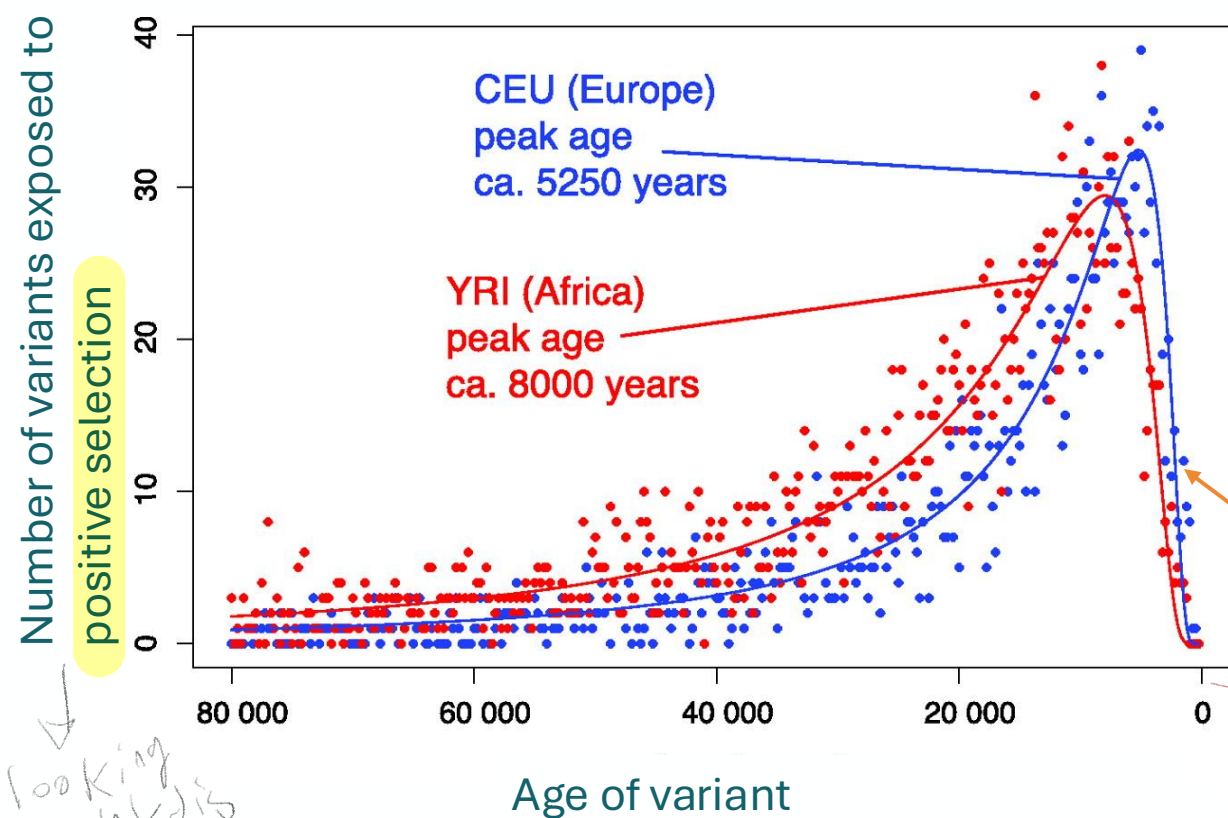
Positive mutations (beneficial alleles) are rare and tend not to stay "segregating" very long.

- If strongly beneficial: sweep to fixation
- If weakly beneficial, they are hard to distinguish from drift or weakly deleterious variants with limited data.

Hence nearly neutral



Rate of adaptive evolution was not neutral across recent human evolutionary history: peak towards the present

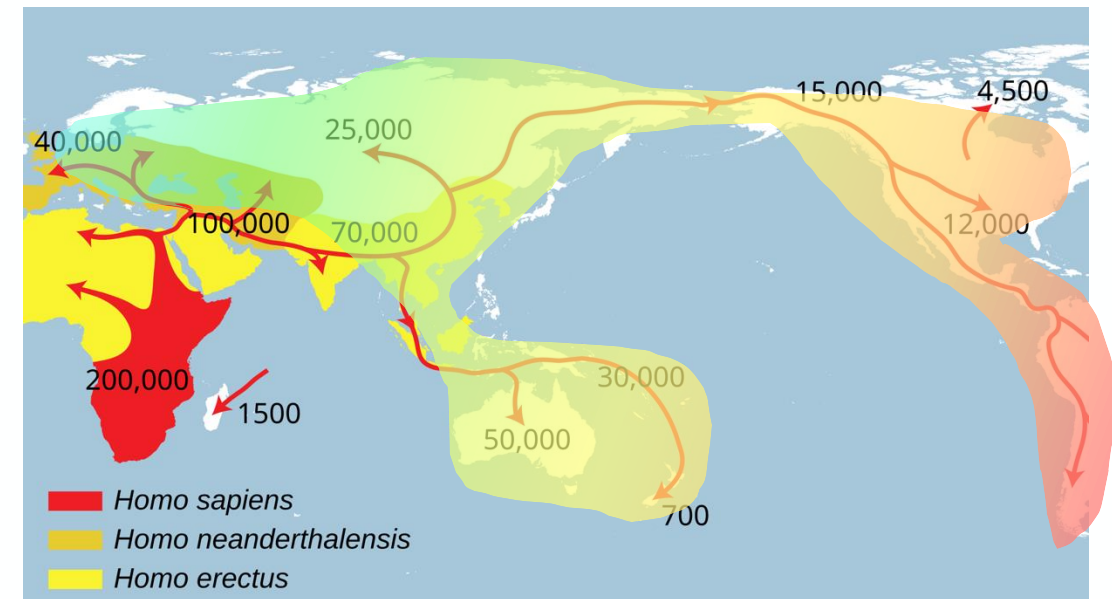
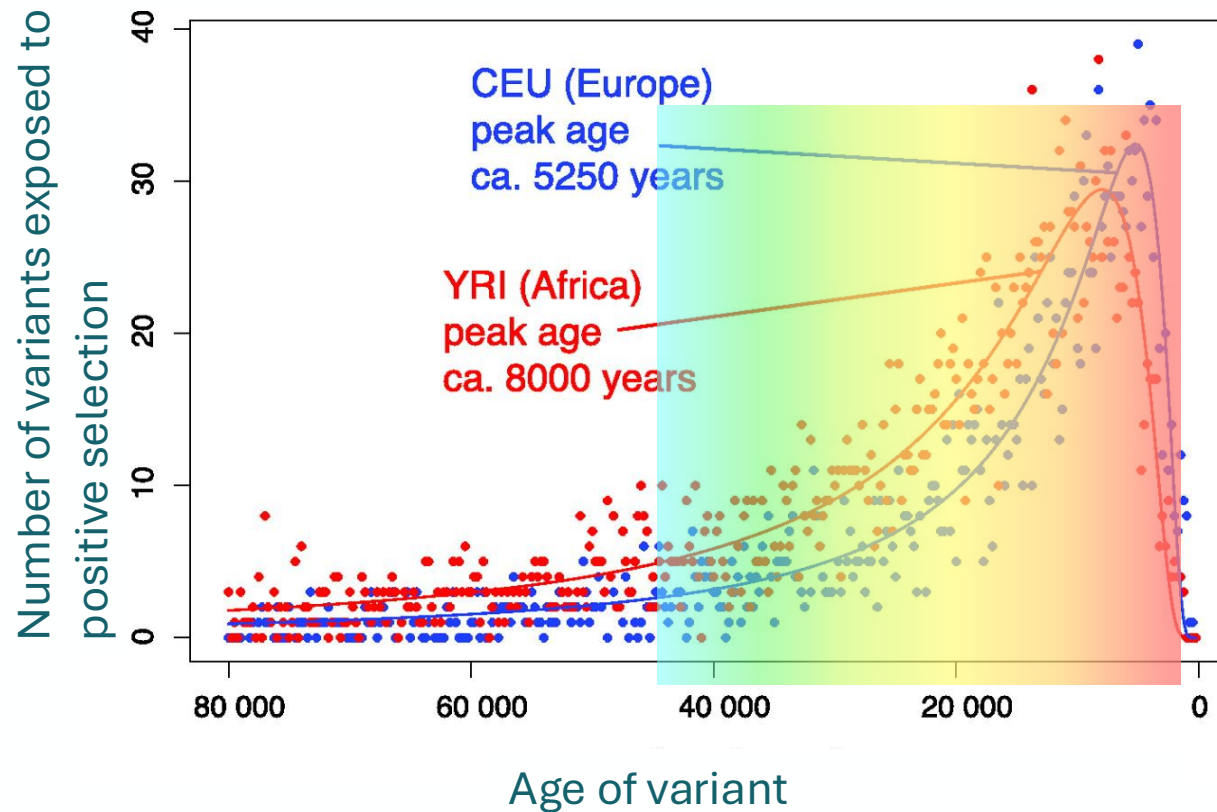


Artifacts of study designs
novel variants that haven't had time for selection yet

“It is sometimes claimed that the pace of human evolution should have slowed as cultural adaptation supplanted genetic adaptation. The high empirical number of recent adaptive variants would seem sufficient to refute this claim...”

Ignore recently appeared variants (until the end of the lecture)

Rate of adaptive evolution was not neutral across recent human evolutionary history: peak towards the present



But maybe these were generally adaptive variants (useful for everyone)... how **locally adapted** are modern humans?

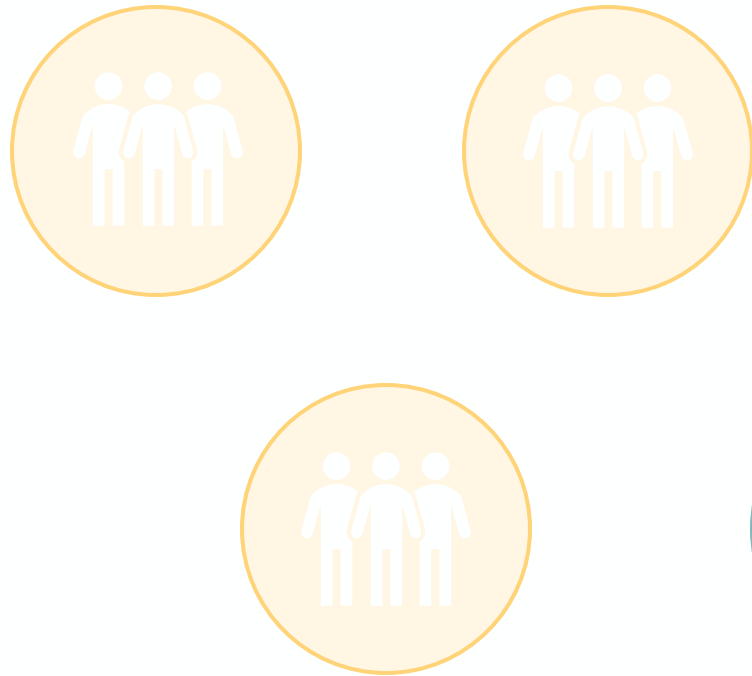
Local adaptation

1. How does it happen? |
2. Selective pressures |
3. Adaptive variation

Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

Environment 1



Environment 2



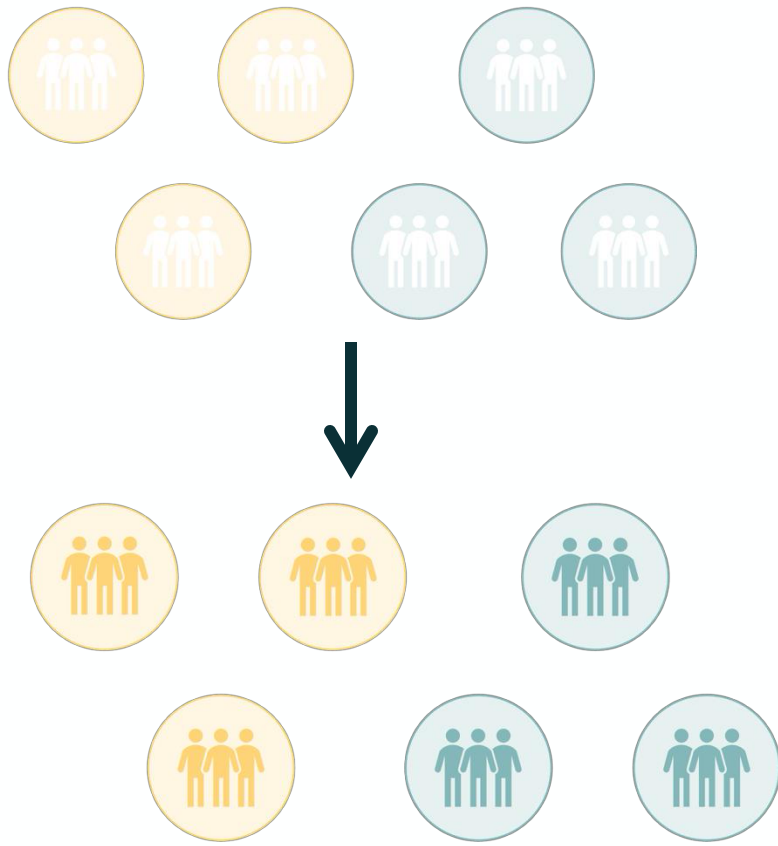
Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Local adaptation

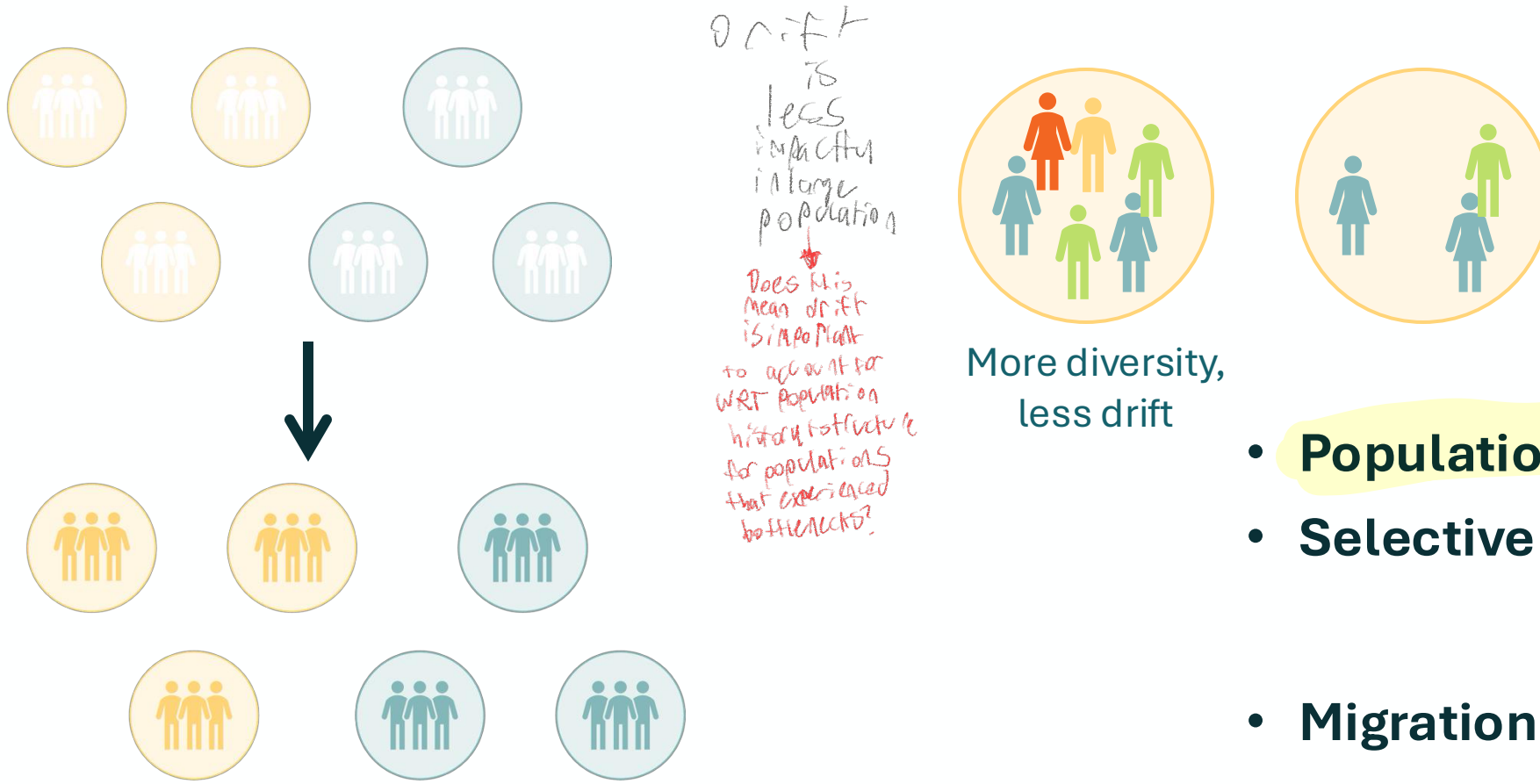
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



- Population size:
- Selective pressure:
- Migration:
- Time:

Local adaptation

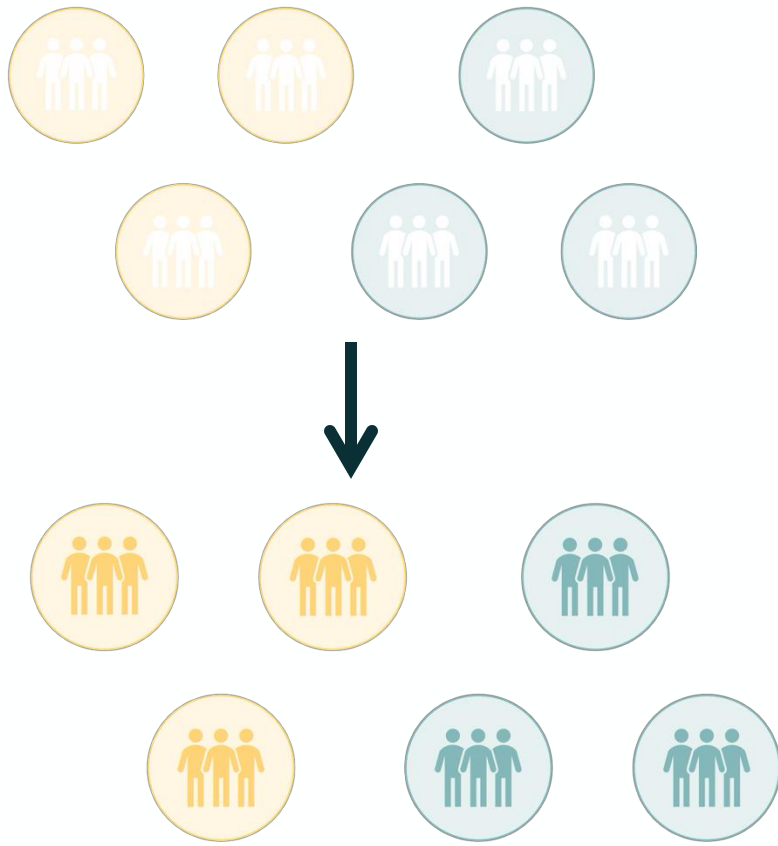
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



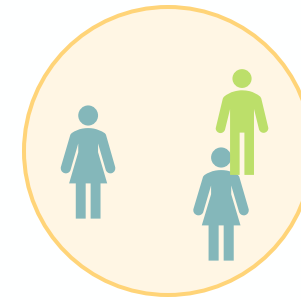
- **Population size:**
- **Selective pressure:**
- **Migration:**
- **Time:**

Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



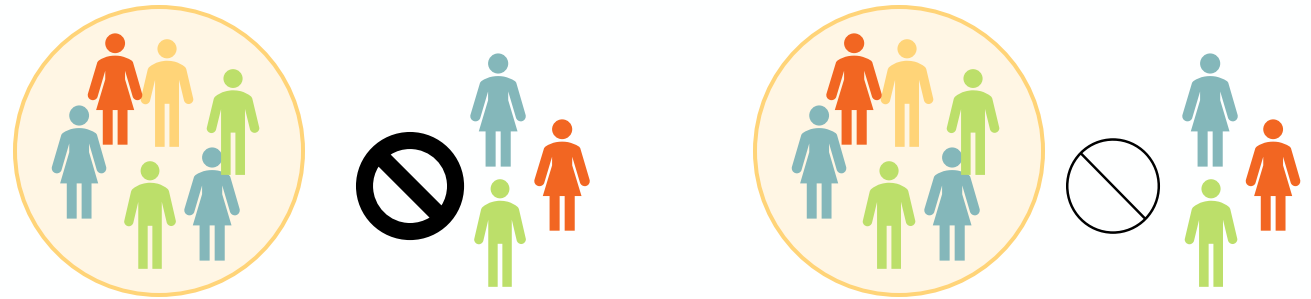
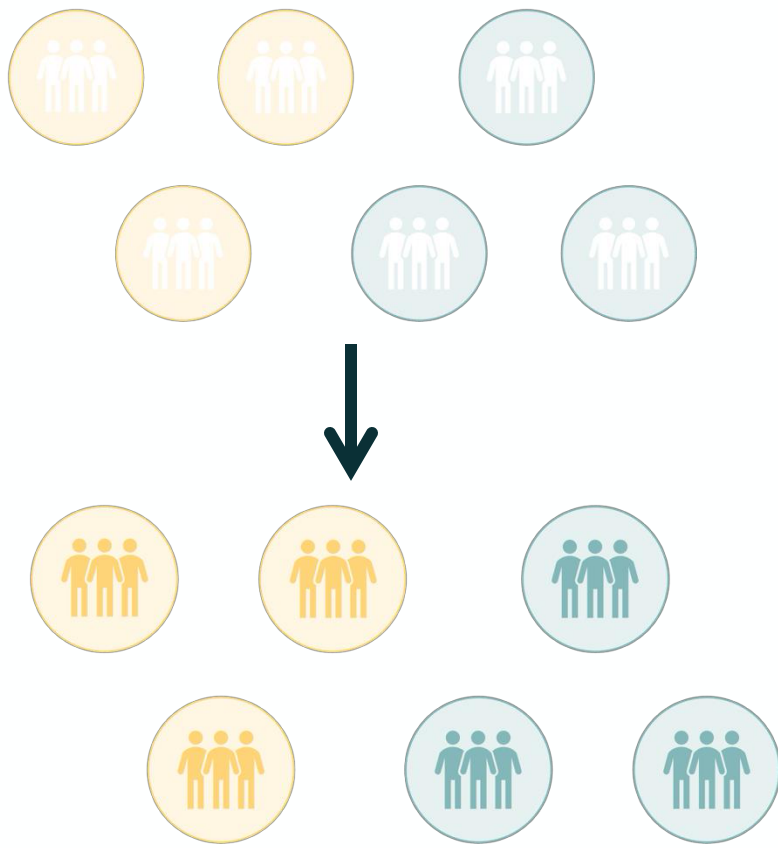
More diversity,
less drift



- **Population size:** Large
- **Selective pressure:**
- **Migration:**
- **Time:**

Local adaptation

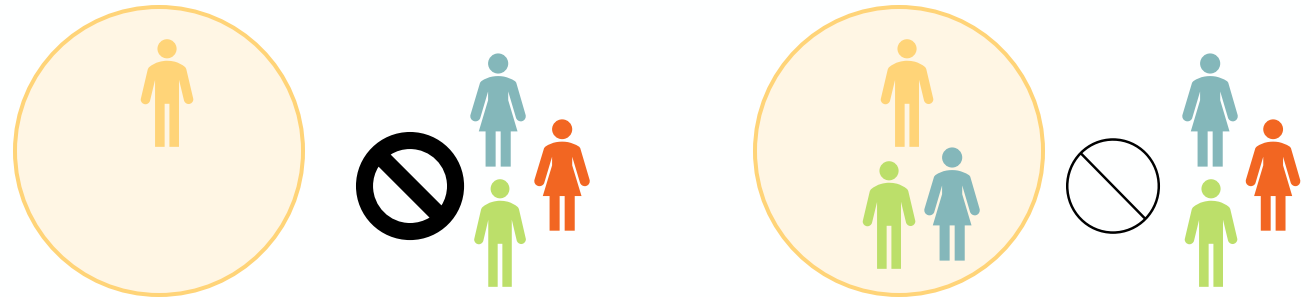
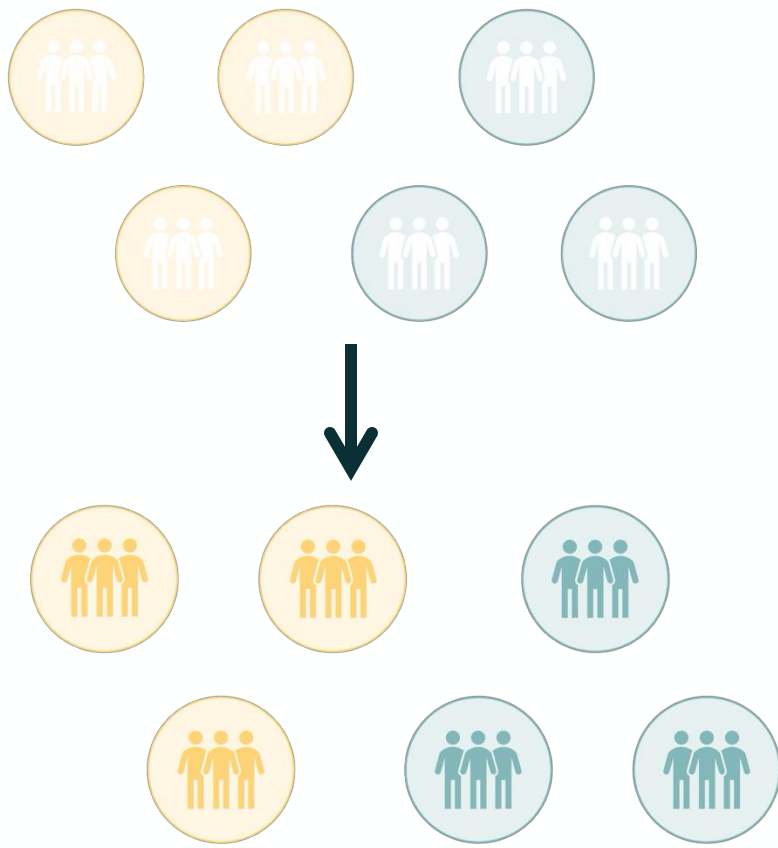
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



- **Population size:** Large
- **Selective pressure:**
- **Migration:**
- **Time:**

Local adaptation

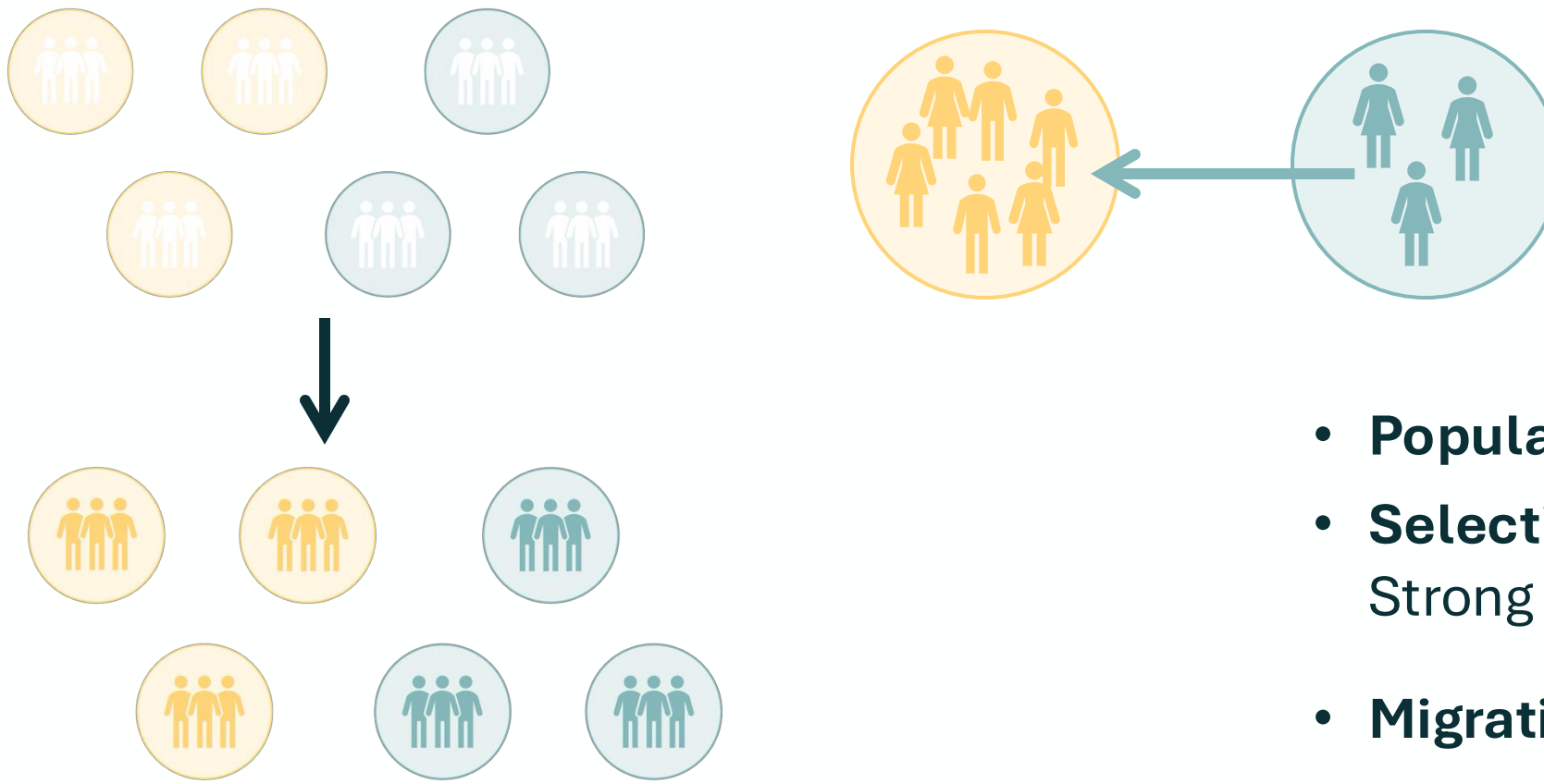
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



- **Population size:** Large
- **Selective pressure:** Strong (& consistent)
- **Migration:**
- **Time:**

Local adaptation

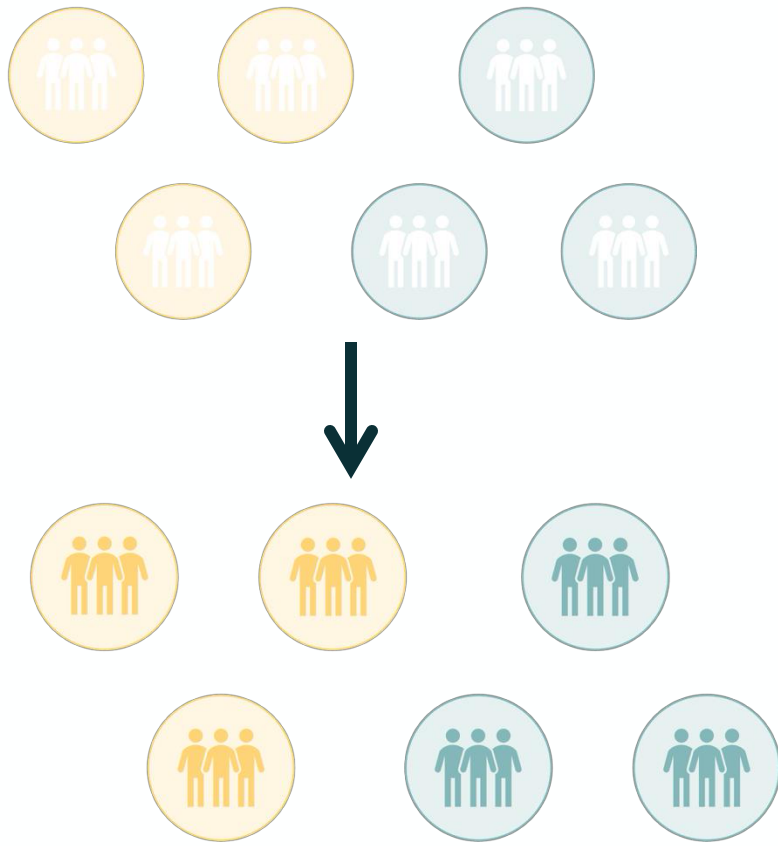
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



- **Population size:** Large
- **Selective pressure:** Strong & consistent
- **Migration:**
- **Time:**

Local adaptation

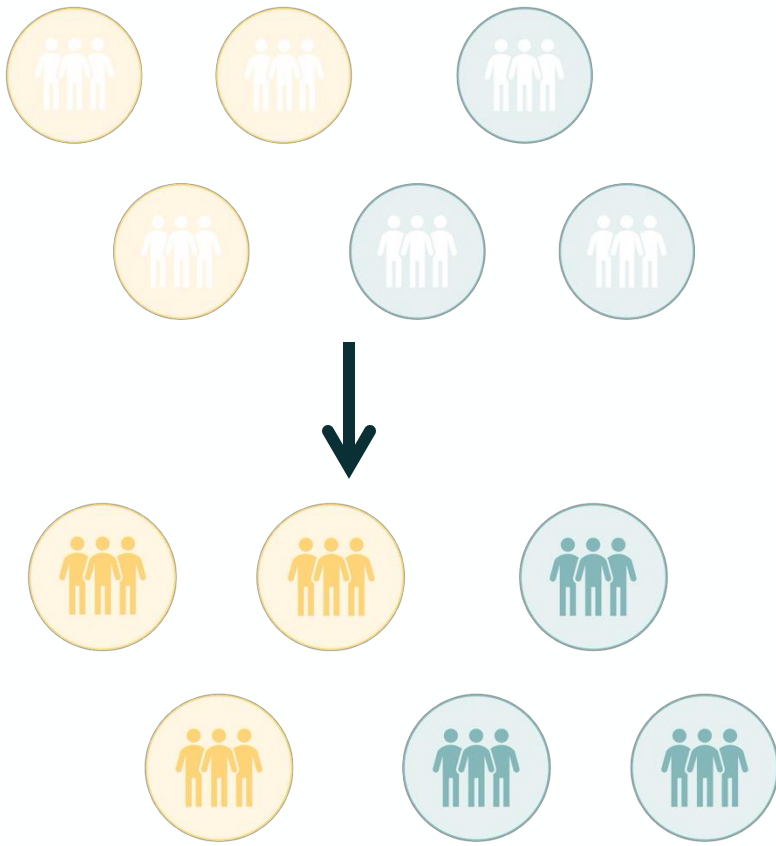
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



- **Population size:** Large
- **Selective pressure:** Strong & consistent
- **Migration:** low levels
- **Time:**

Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

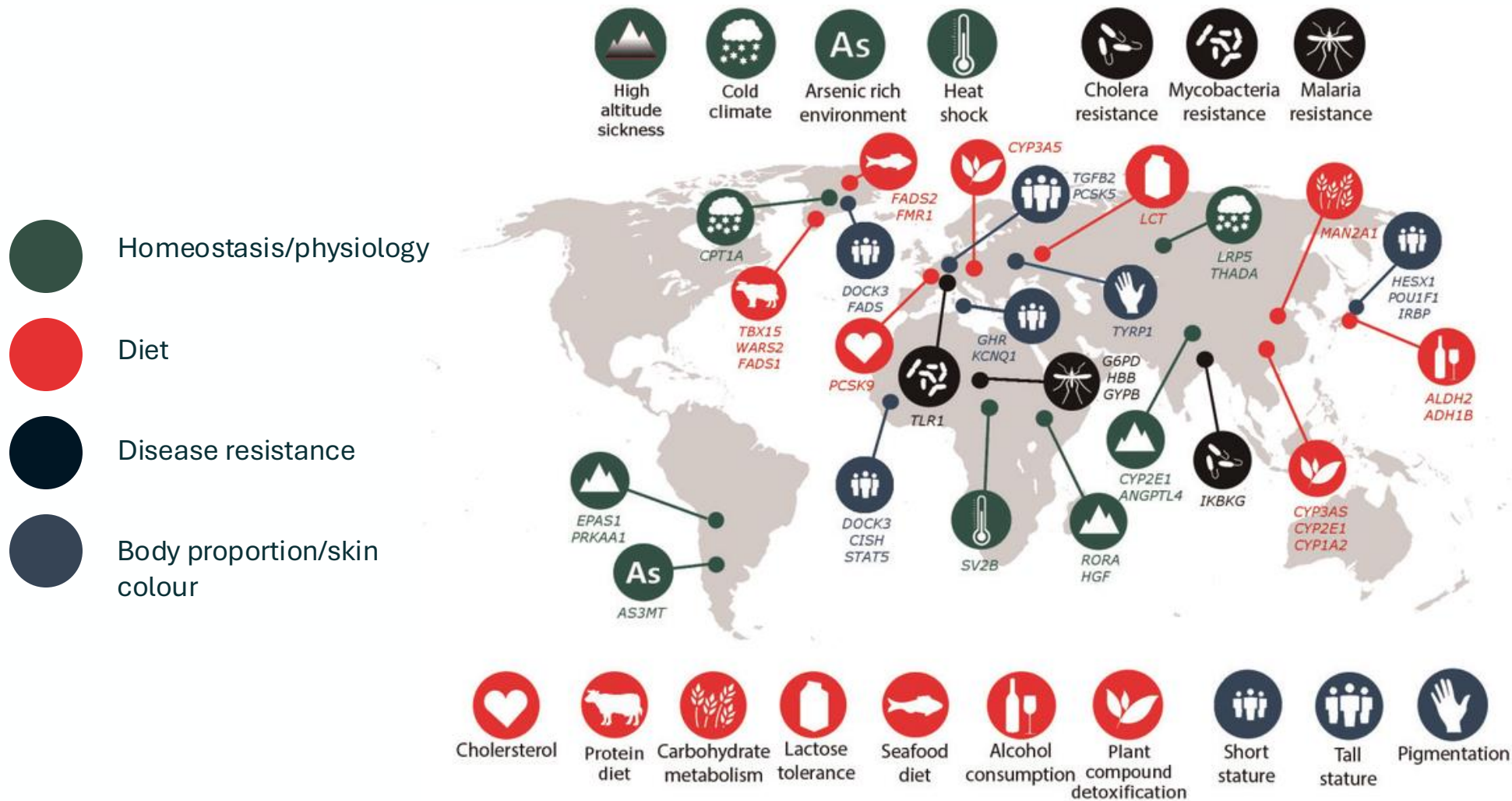


*To allow for local adaptation,
you need...*

- **Population size:** Large
- **Selective pressure:** Strong & consistent
- **Migration:** low levels
- **Time:** Long

Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Hey I know
that guy!

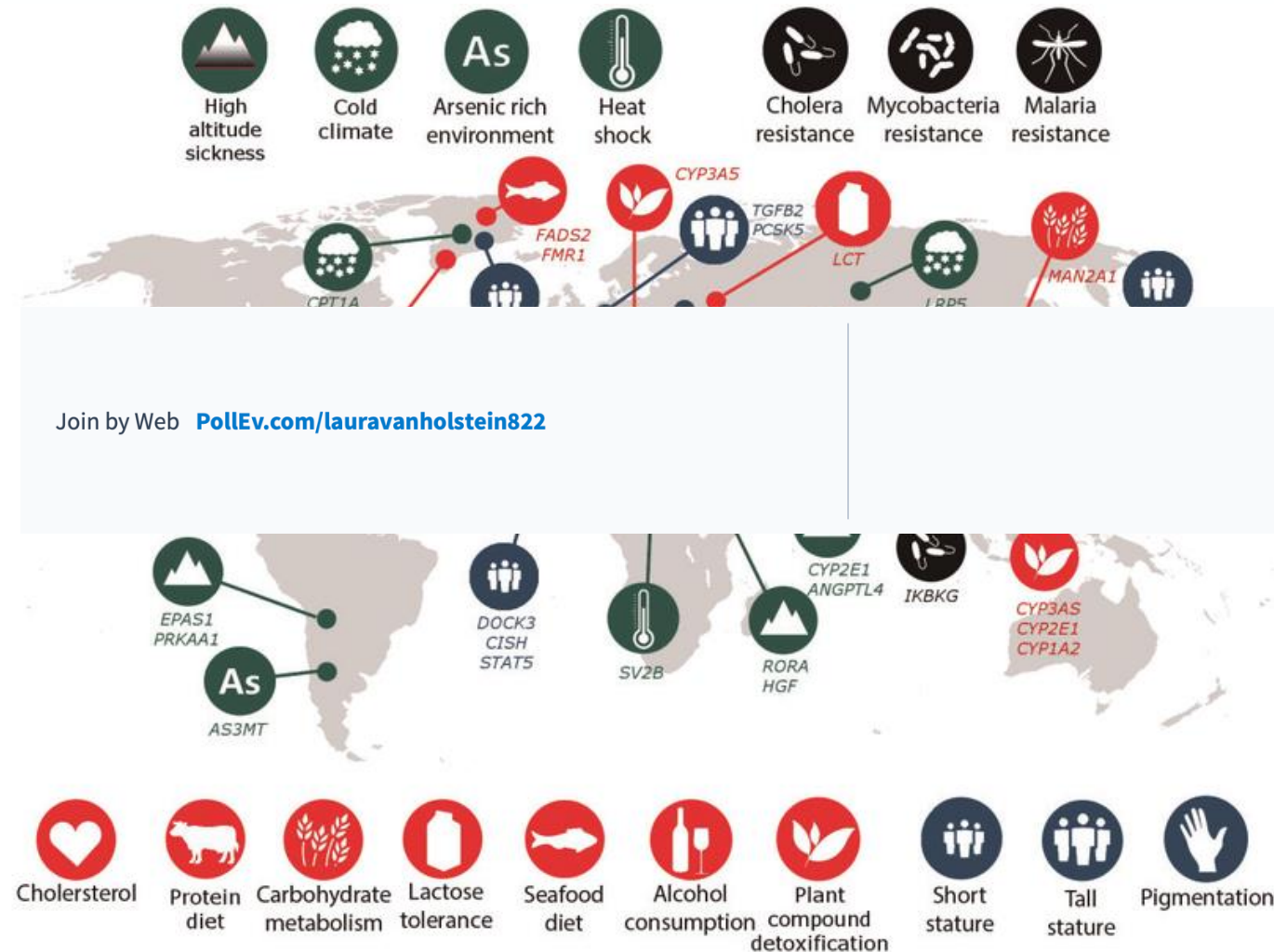
Elhaik et al. (2017),
based on Fan et al.
(2016)

Local adaptation

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

Selective factor

- Ecology
- Energy extraction
↳ Nutrition
- Disease mortality
- Various



Join by Web PollEv.com/lauravanholstein822



Elhaik et al. (2017),
based on Fan et al.
(2016)

Dairy consumption



Energy extraction

1. How does it happen?
2. Selective pressures
3. Adaptive variation



Childhood: lactase enzyme; **can digest lactose**

Adult: lactase non-persistence (LNP); **cannot digest lactose**



Adult: lactase enzyme (LP); **can digest lactose**



Lactose tolerance – caused by lactase *persistence* (LP) – is at high frequency in Europe, and areas of Africa, the Middle East, and South Asia:

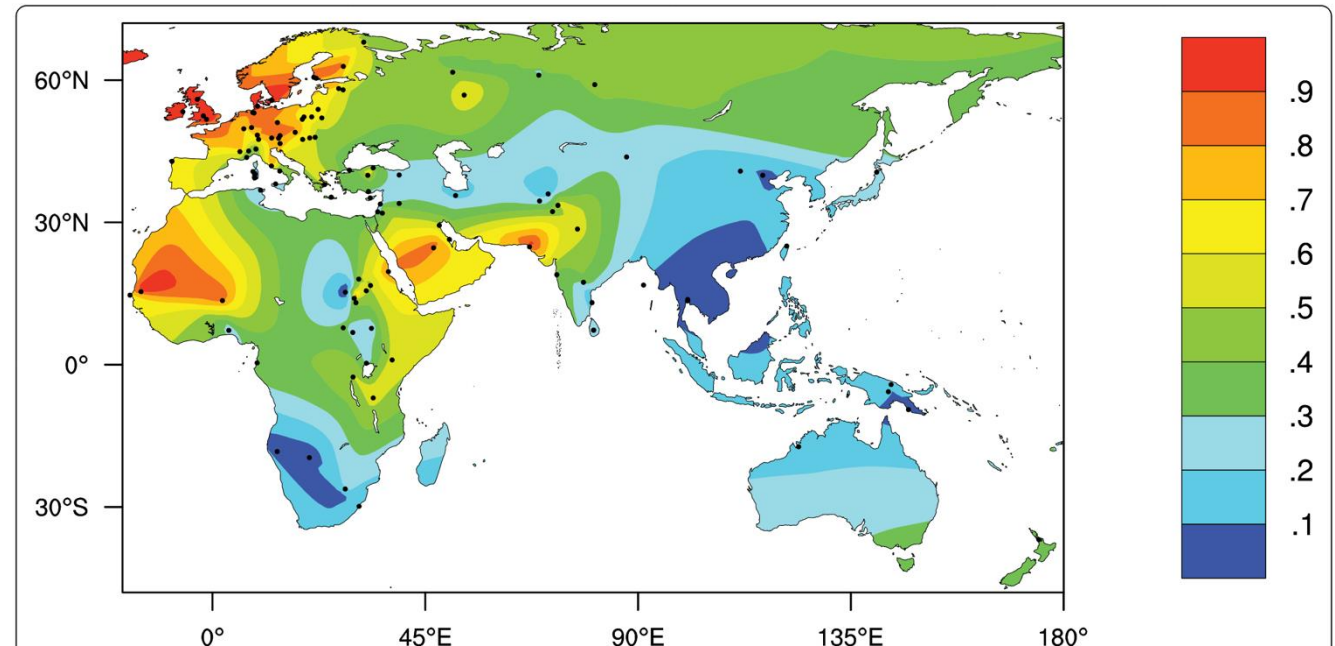


Figure 1 Interpolated map of Old World LP phenotype frequencies. Dots represent collection locations. Colours and colour key show the frequencies of the LP phenotype estimated by surface interpolation.

Dairy consumption



1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



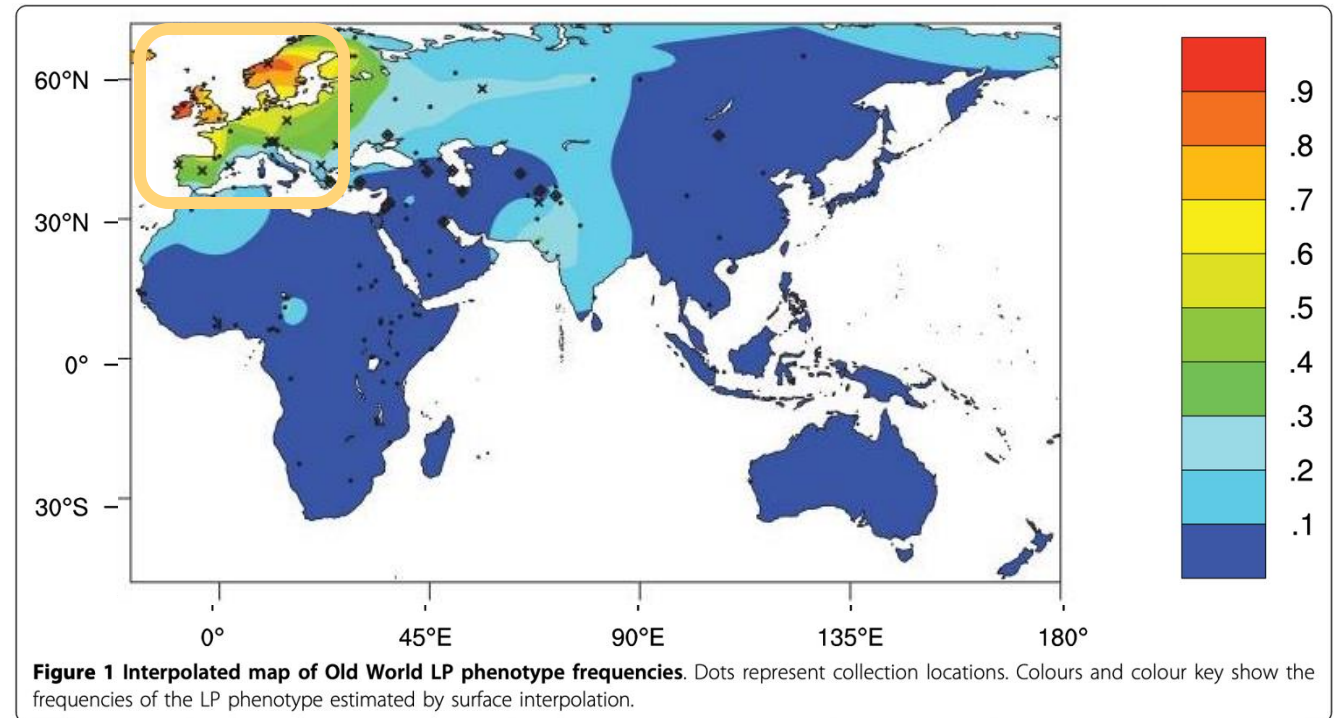
Lactase persistence in people of European ancestry is caused by a mutation in the MCM6 gene that regulates lactase production.

Some of the **strongest signals** of selection in the human genome

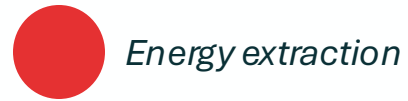
↳ For lactase persistence



Rates of MCM6 for LP



Dairy consumption

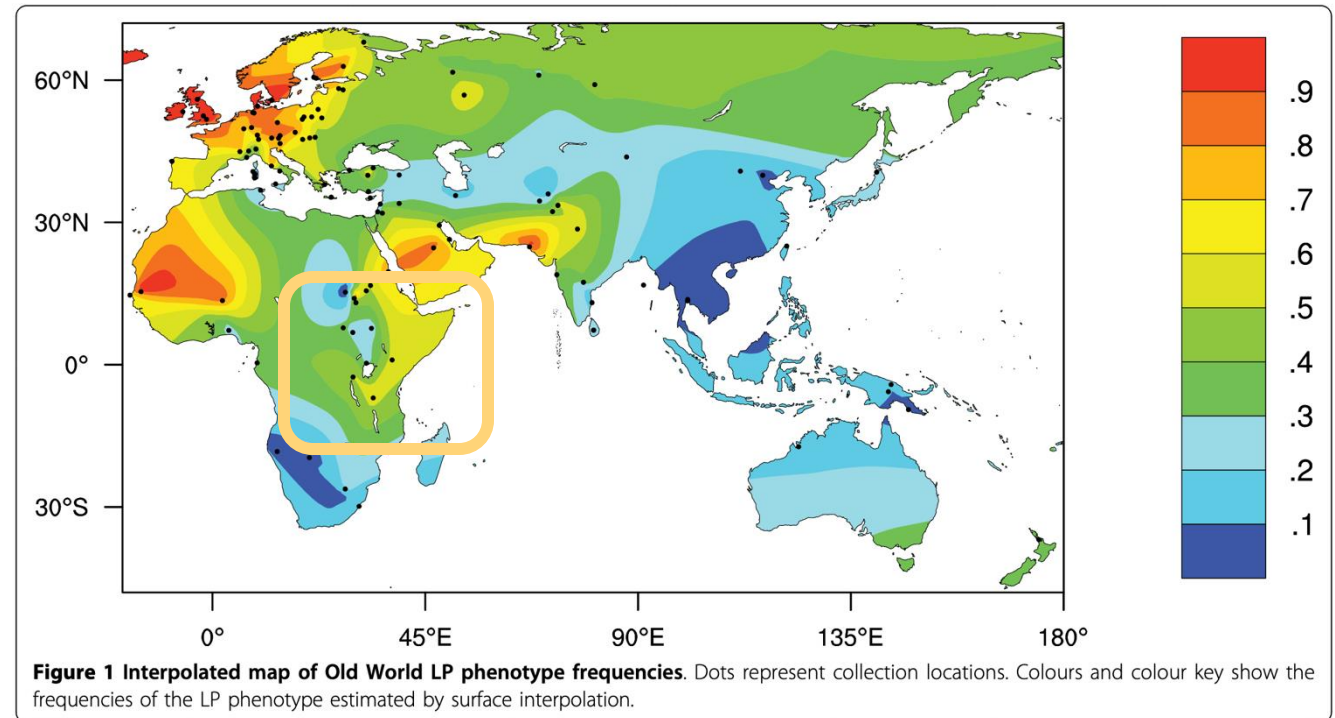


1. How does it happen?
2. Selective pressures
3. Adaptive variation



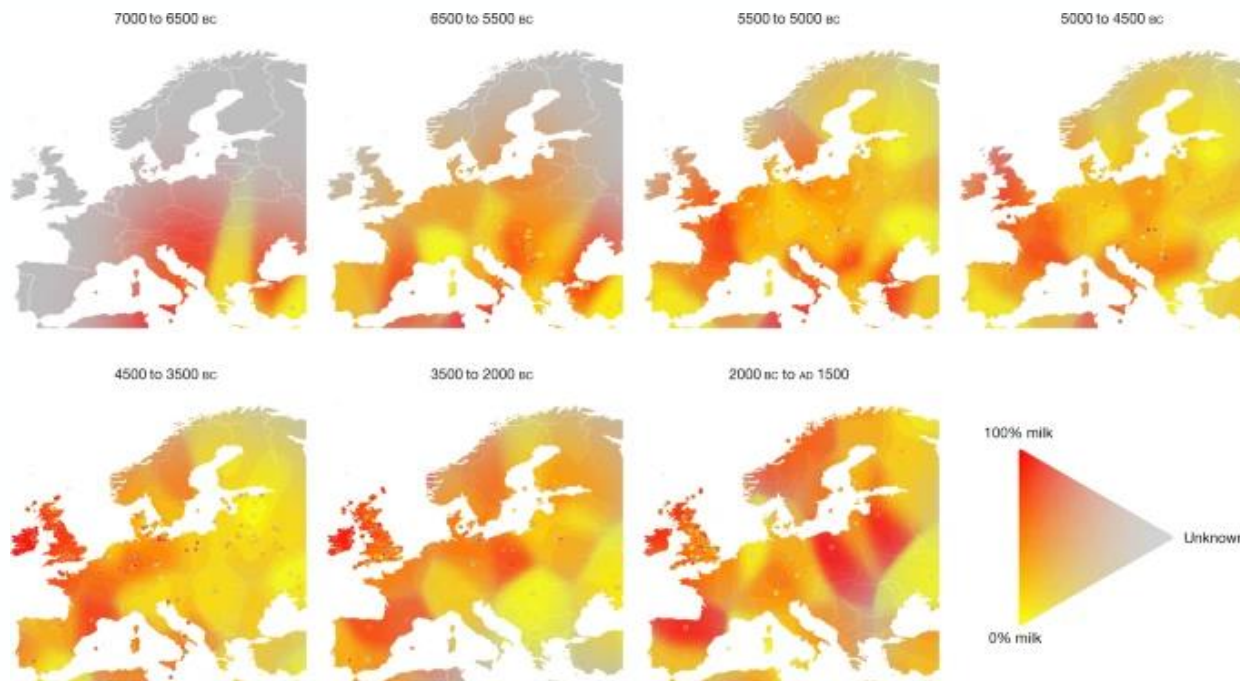
Lactase persistence in people of African ancestry is caused by SNPs located within 100 base pairs of the European variant

> **Convergent evolution**



Dairy consumption *Energy extraction*

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Spread of dairy consumption
(measured in potsherds with dairy fat)



Neolithic dairying

Dairy consumption *Energy extraction*

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Dairy consumption Energy extraction

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

- **Calcium assimilation hypothesis**

Milk supplements low vitamin D diets in regions with reduced ultraviolet B radiation

However, this hypothesis does not explain the inferred high selection strengths in lower latitude regions, such as Africa, the Middle East, southern Europe and southern Asia!

- **Pathogen-free fluid hypothesis** *Selection for being able to get hydration from source without pathogens*
Milk as a source of relatively pathogen-free fluid; likely exceptionally important in arid regions

- **Nutritional benefits**

Carbohydrates as well as fat, protein and calcium

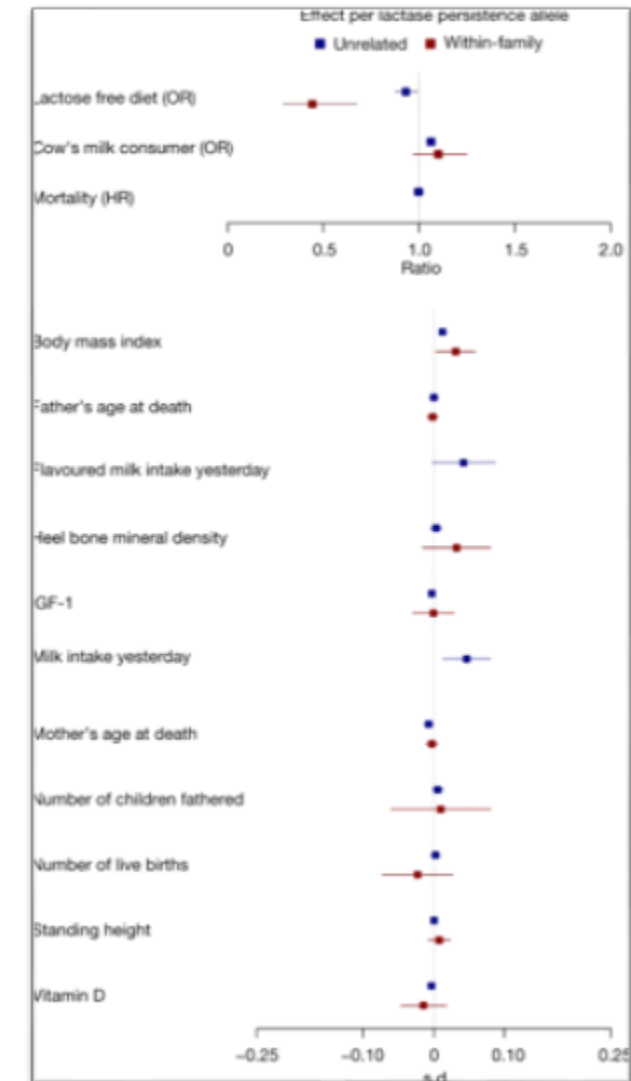


Neolithic dairying

Dairy consumption *Energy extraction*

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

- Calcium assimilation hypothesis
- Pathogen-free fluid hypothesis
- Nutritional benefits



Dairy consumption ● *Energy extraction*

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

• ~~Calcium assimilation hypothesis~~

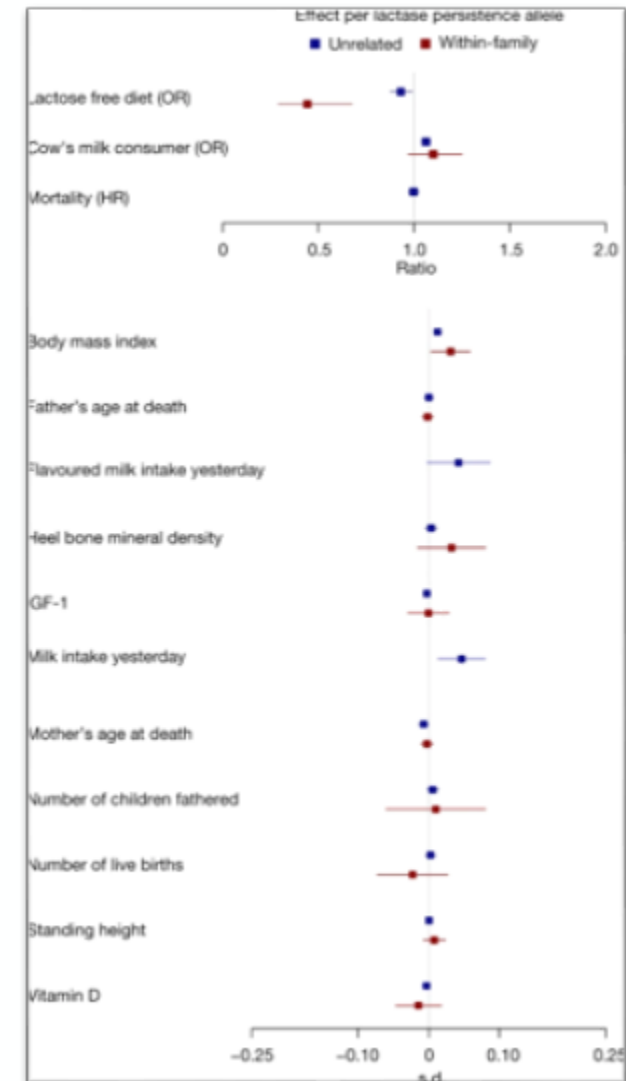
- No lower bone mineral density in LNP
- No higher vitamin D

The traits you would expect to see if this hypothesis was false are absent

• Pathogen-free fluid hypothesis

• Nutritional benefits

Testing disproves this hypothesis as reason for selection for LP



Dairy consumption Energy extraction

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

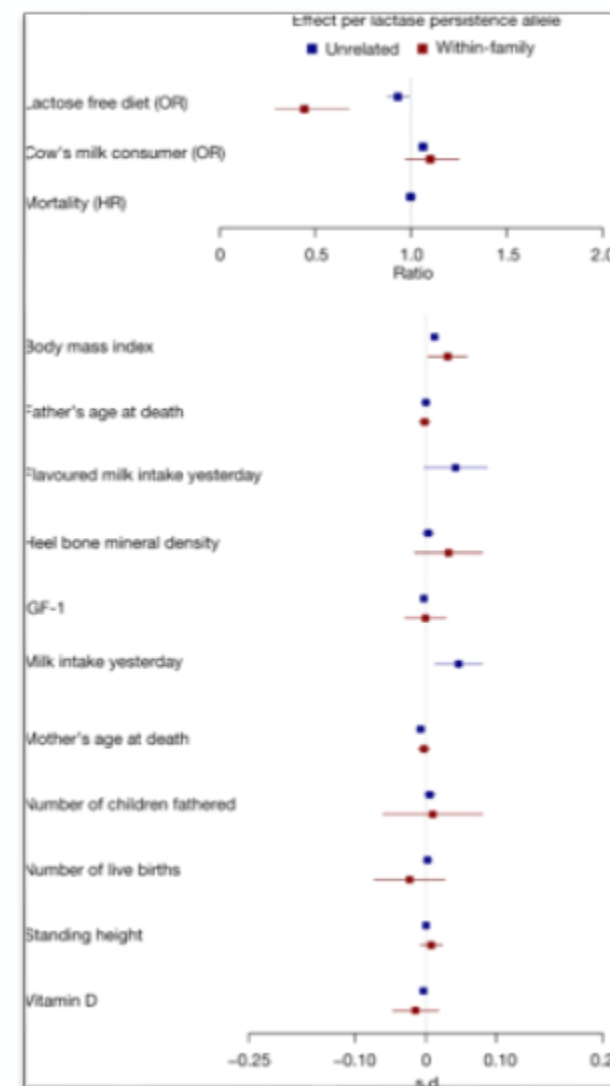
• ~~Calcium assimilation hypothesis~~

- No lower bone mineral density in LNP
- No higher vitamin D

• ~~Pathogen-free fluid hypothesis~~

- No higher mortality in LNP – *No higher mortality for people who can't digest milk*

• Nutritional benefits



Dairy consumption Energy extraction

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

• ~~Calcium assimilation hypothesis~~

- No lower bone mineral density in LNP
- No higher vitamin D

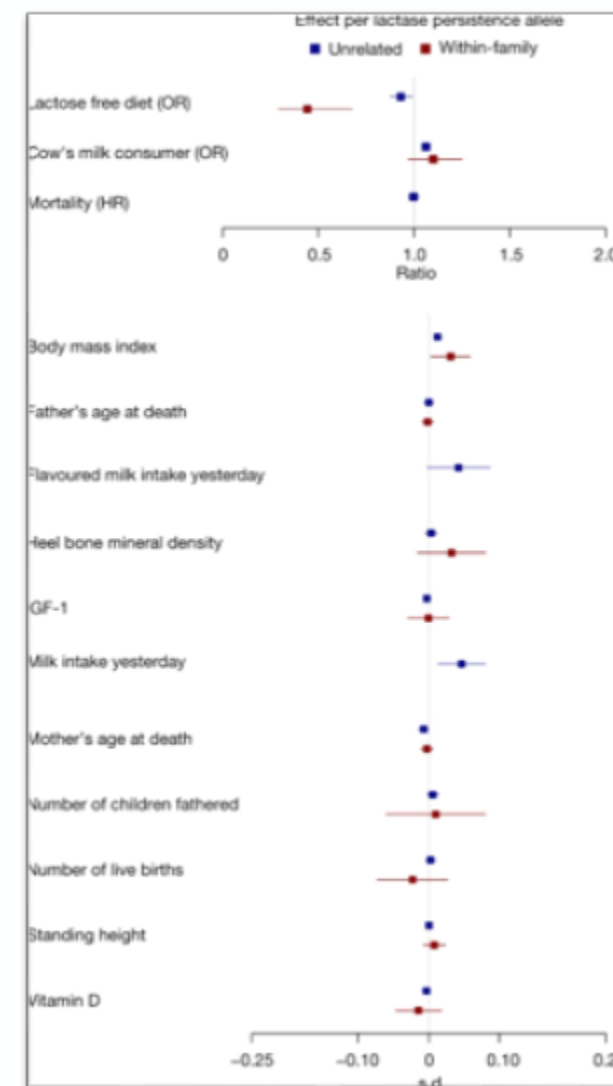
• ~~Pathogen-free fluid hypothesis~~

- No higher mortality in LNP

• ~~Nutritional benefits~~

- LNP individuals drink cow's milk & don't follow a lactose free diet

In addition, no lower fitness in LNP males and females
So how is LNP *maladaptive*, and/or how is LP adaptive?



Dairy consumption



Energy extraction



Disease mortality



Famine

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Increased population and settlement density, proximity to animals, crop failure, famine and population collapse, general poor hygiene and sanitation

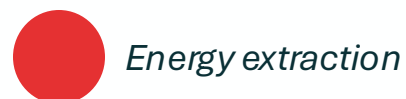
*Leading to...
proven: a microbiological
record that identifies lower
sanitation?*

Crisis:

LP selection pressures (negative selection for LNP, positive selection for LP) **greater during times of subsistence instability (famine)**

- Normally, LNP not fatal.
- In famine, maybe: lactose-induced diarrhoea can shift from an inconvenient to a fatal condition in severely malnourished individuals and high-lactose (unfermented) milk products are more likely to be consumed when other food sources have been exhausted

Dairy consumption



Energy extraction



Disease mortality



Famine

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Increased population and settlement density, proximity to animals, crop failure, famine and population collapse, general poor hygiene and sanitation

Increased infectious disease loads, particularly zoonoses

Crisis:

LP selection pressures (negative selection for LNP, positive selection for LP) greater during times of subsistence instability (famine)

- Normally, LNP not fatal.
- In famine, maybe: lactose-induced diarrhoea can shift from an inconvenient to a fatal condition in severely malnourished individuals and high-lactose (unfermented) milk products are more likely to be consumed when other food sources have been exhausted

Chronic:

LP selection pressures (negative selection for LNP, positive selection for LP) increase with increased infection exposure

- Normally, LNP not fatal.
- Mortality and morbidity due to pathogen exposure would have been amplified by the otherwise minor health effects of LNP in individuals consuming milk—particularly diarrhoea—due to fluid loss and other gut disturbances, leading to enhanced selection for LP

Dairy consumption



Energy extraction

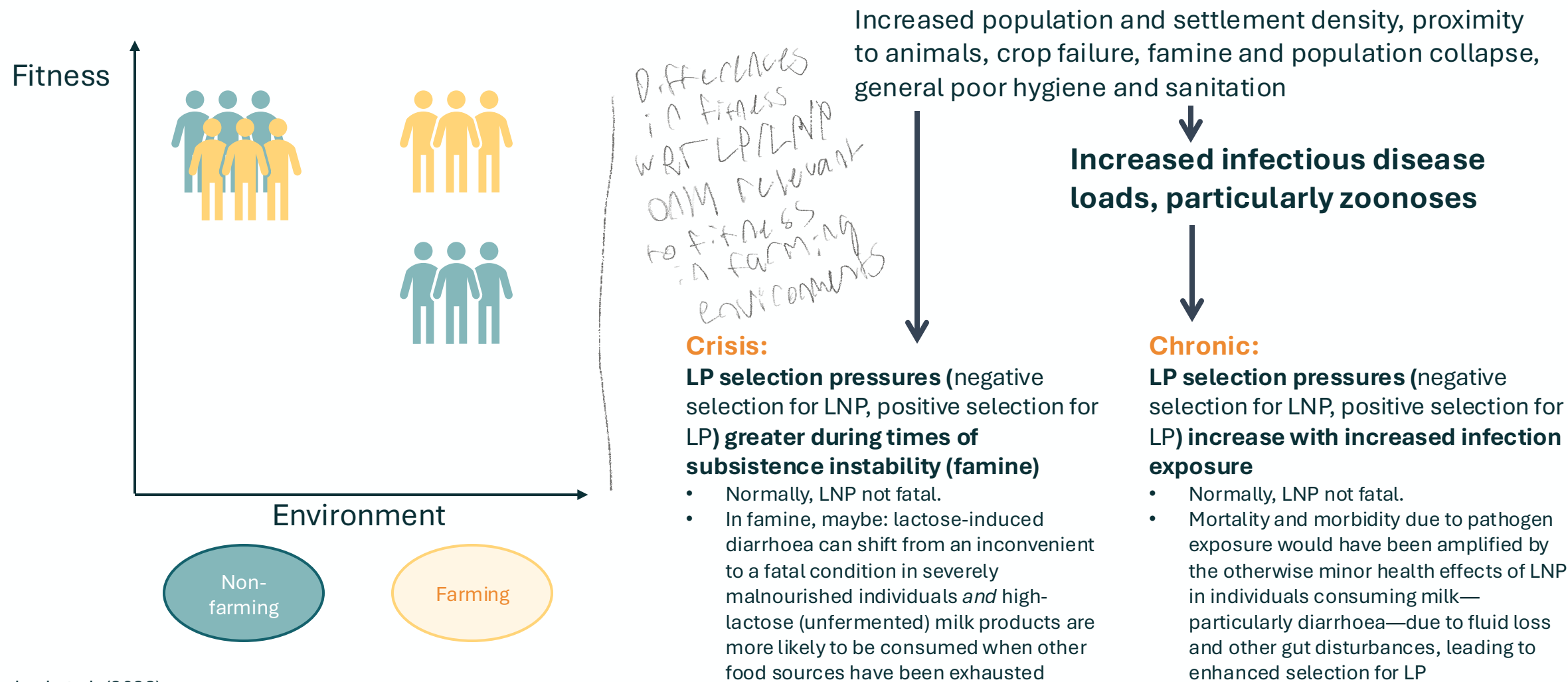


Disease mortality

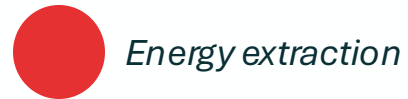


Famine

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



Dairy consumption



Energy extraction



Disease mortality



Famine

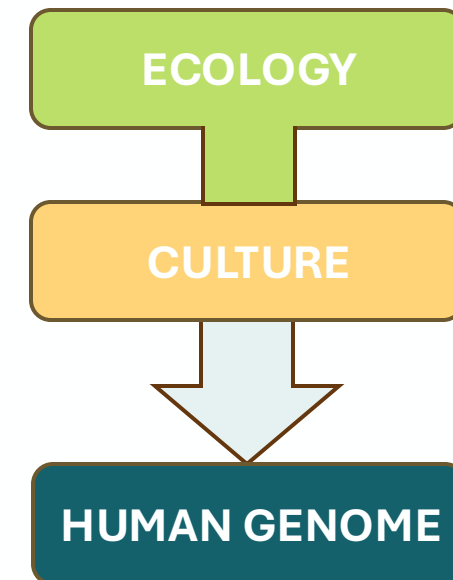
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

Niche construction is the process by which organisms construct (components of) their local environment in ways **that introduce novel selection pressures**.

Col FO adl
adaptation is
what influenced
selective
pressures

“Lactase persistence is one of the clearest examples of **niche construction** in humans.”

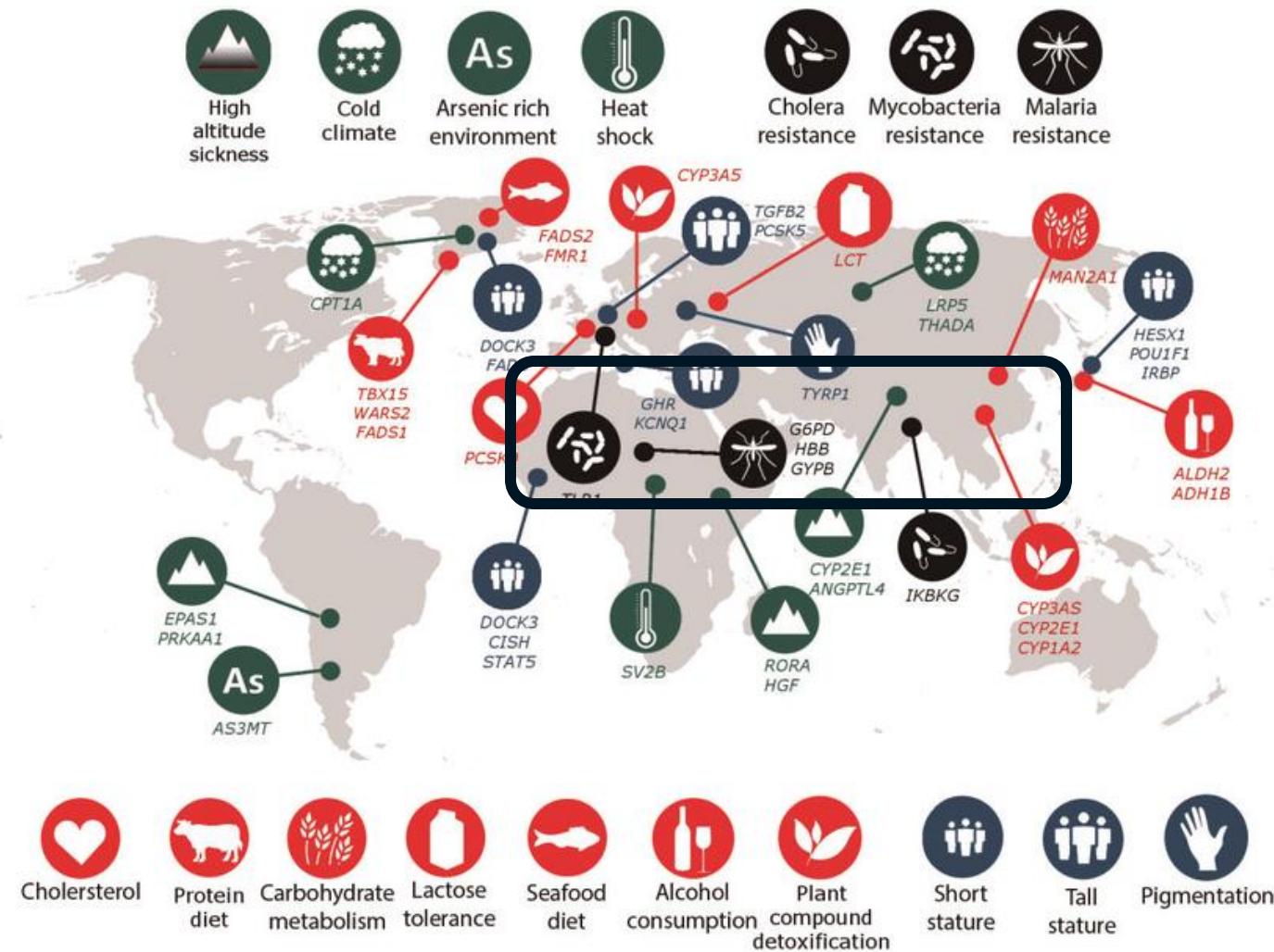
Gerbault et al. (2011)



Endemic pathogens



1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



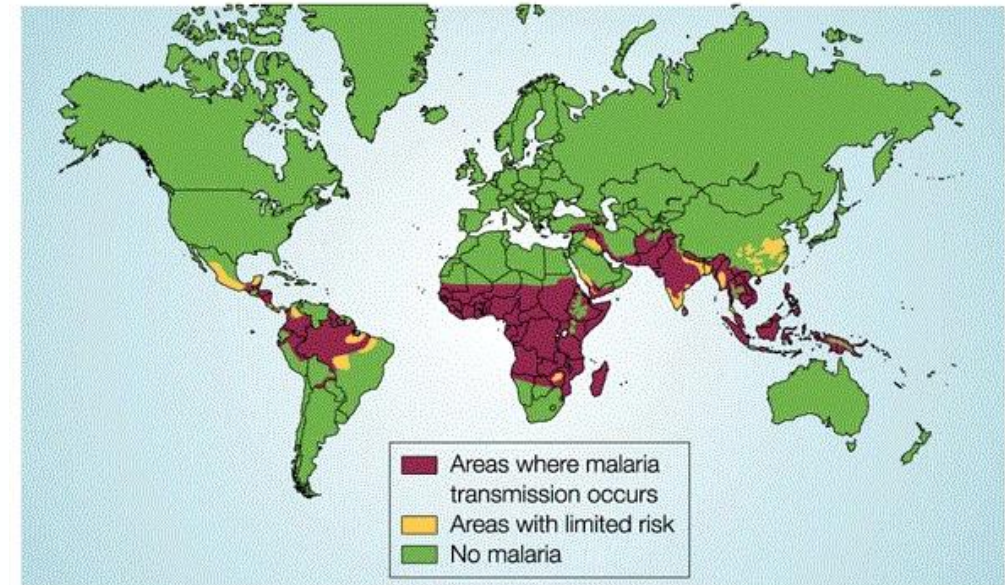
Elhaik et al. (2017),
based on Fan et al.
(2016)

Endemic pathogens Disease mortality

1. How does it happen?
2. Selective pressures
3. Adaptive variation

Tropical rainforests have exceptional **high levels of endemic pathogens**

Expected to exert **very strong** selection, especially if these cause pre-reproductive mortality



Nature Reviews | Microbiology

Malaria



Disease mortality

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

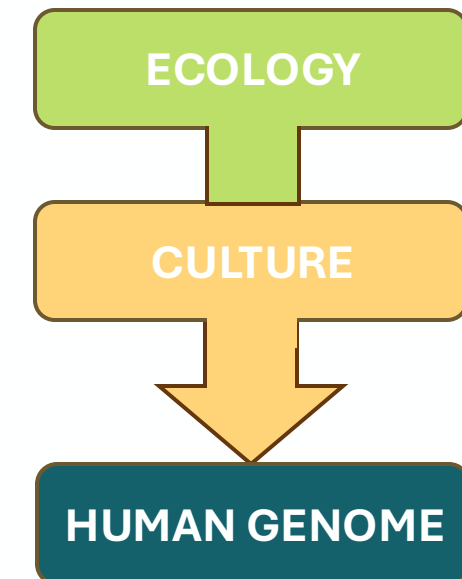
One of the leading causes of child mortality

Selection likely increased with higher population densities and farming in the Neolithic

Many genetic routes to resistance:

- Duffy (ACKR1)
- Sickle cell anaemia (HBB)
- G6PD deficiency (G6PD)
- SE Asian ovalocytosis (SLC4A1)
- CD36
- Gerbich negative (GYPC)
- α and β -Thalassemia (HBA1/HBA2, HBB)
- and more...

→ For malaria's standing water?



Malaria



Disease mortality

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

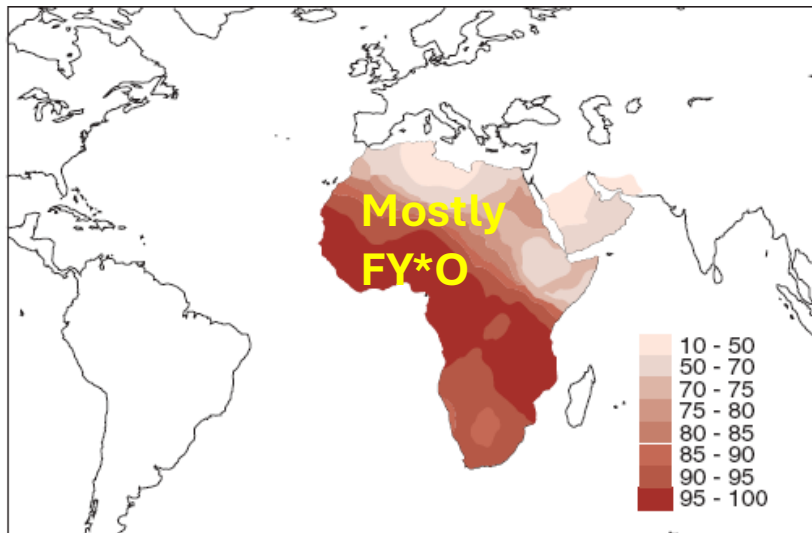
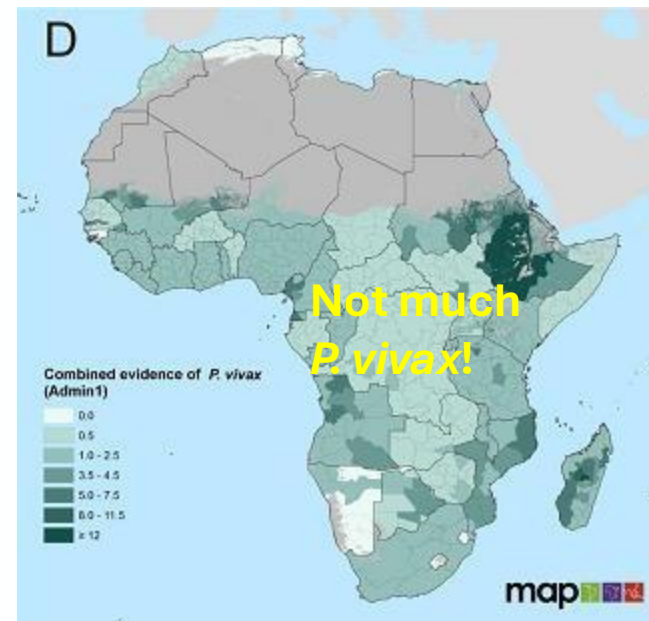


Fig. 5. Extreme population differences in *FY*O* allele frequency. The *FY*O* allele, which confers resistance to *P. vivax* malaria, is prevalent and even fixed in many African populations, but virtually absent outside Africa (38).



- FY gene encodes the Duffy blood antigen system
- **Duffy negative allele (FY*O)** confers resistance against malaria
- High **Duffy negative allele (FY*O)** frequency thought to be due to selection for resistance to *Plasmodium vivax* malaria. Low levels of *P. vivax* in Africa are thought to be related to the Duffy negative mutation!
- But, tradeoff: FY*O individuals are more susceptible to HIV.

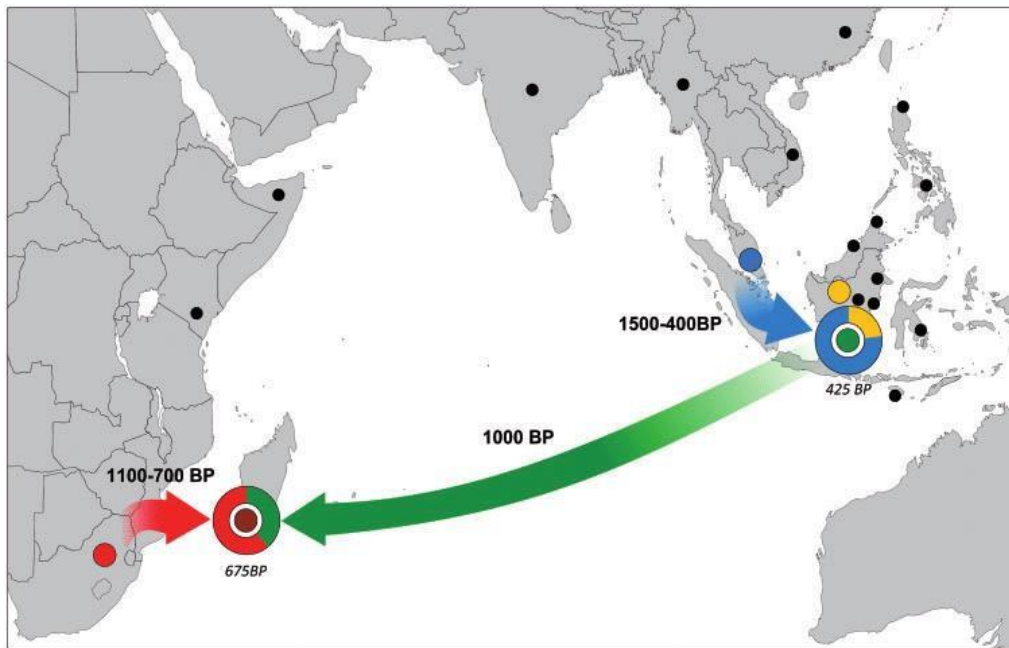
A different
type of malaria
parasite

Malaria



Disease mortality

1. How does it happen? | 2. Selective pressures | 3. Adaptive variation



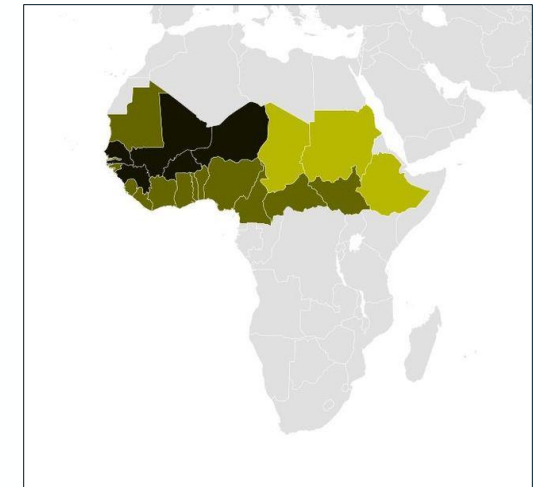
Adaptive admixture at the Duffy locus

Madagascar

- Mixed Asian and African ancestry due to Austronesian migrations
- Excess African ancestry at Duffy locus is consistent with very strong selection!

Fula people of West Africa

- Some Eurasian ancestry
- Excess African ancestry at Duffy locus



↗ Rainforest hunter-gatherers

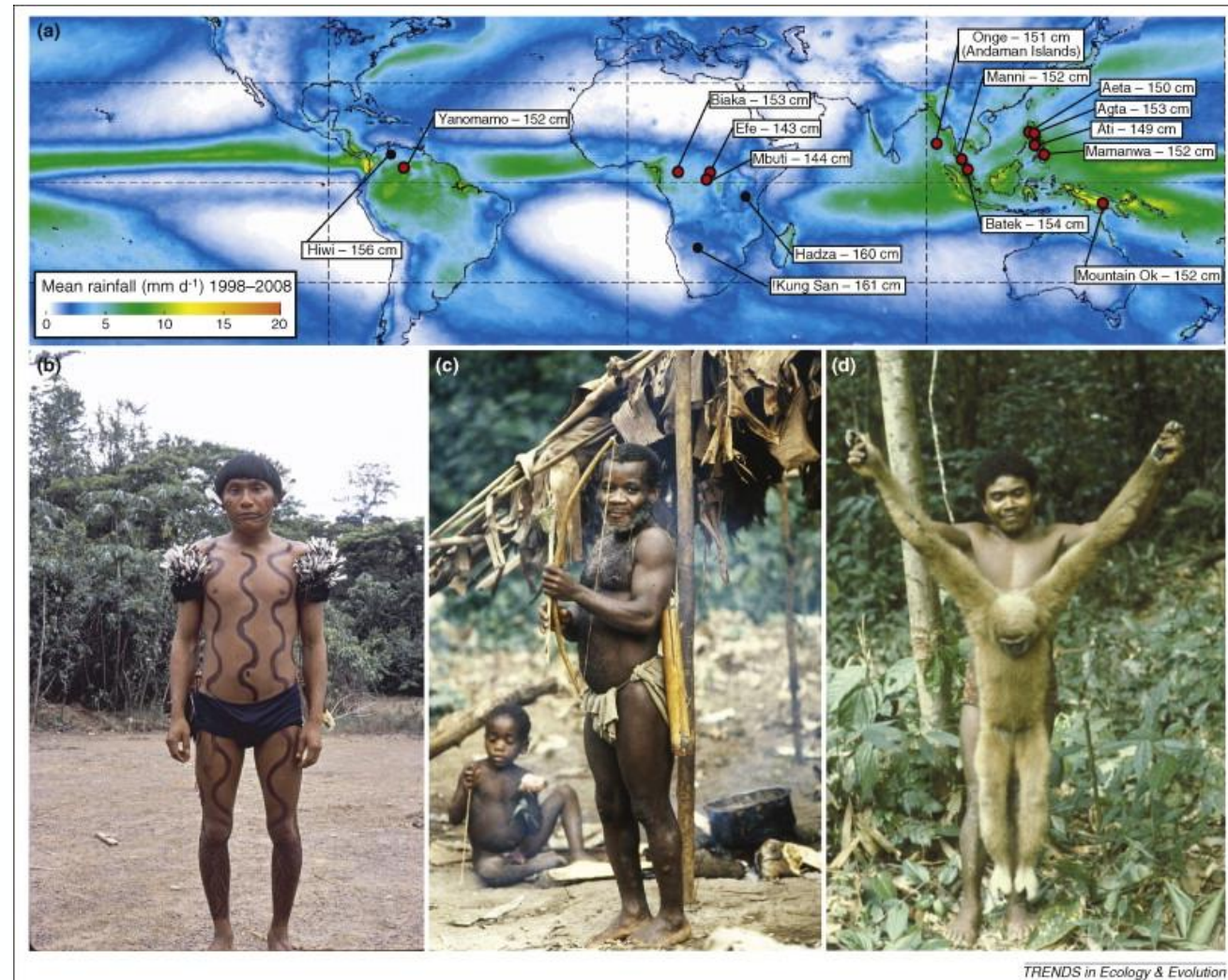
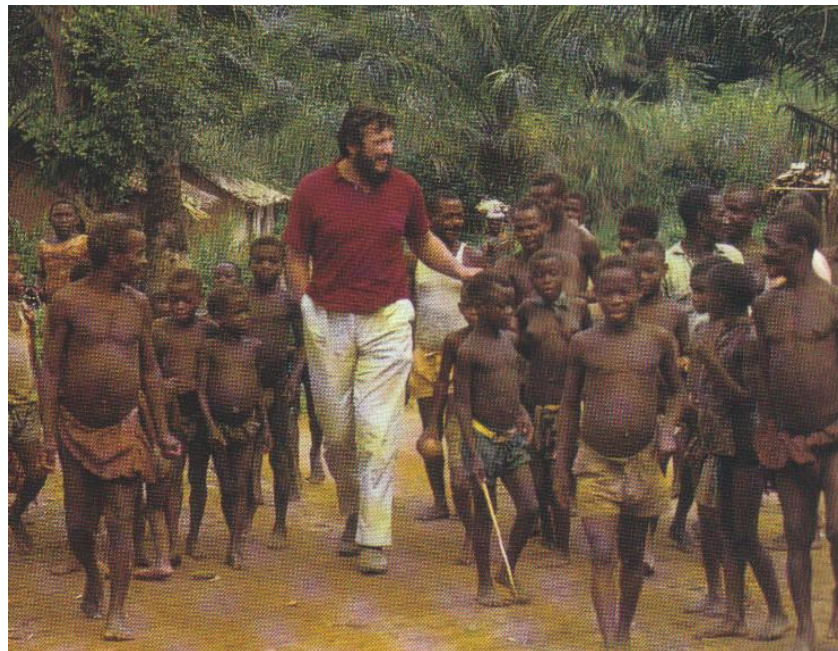
Short stature in RFHG



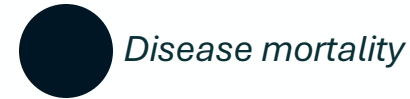
1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

Rainforests:

- High temperature *and* humidity
- Parasites and other pathogens
- 'Pygmy' phenotype: average height of <150cm in adult males



Short stature in RFHG

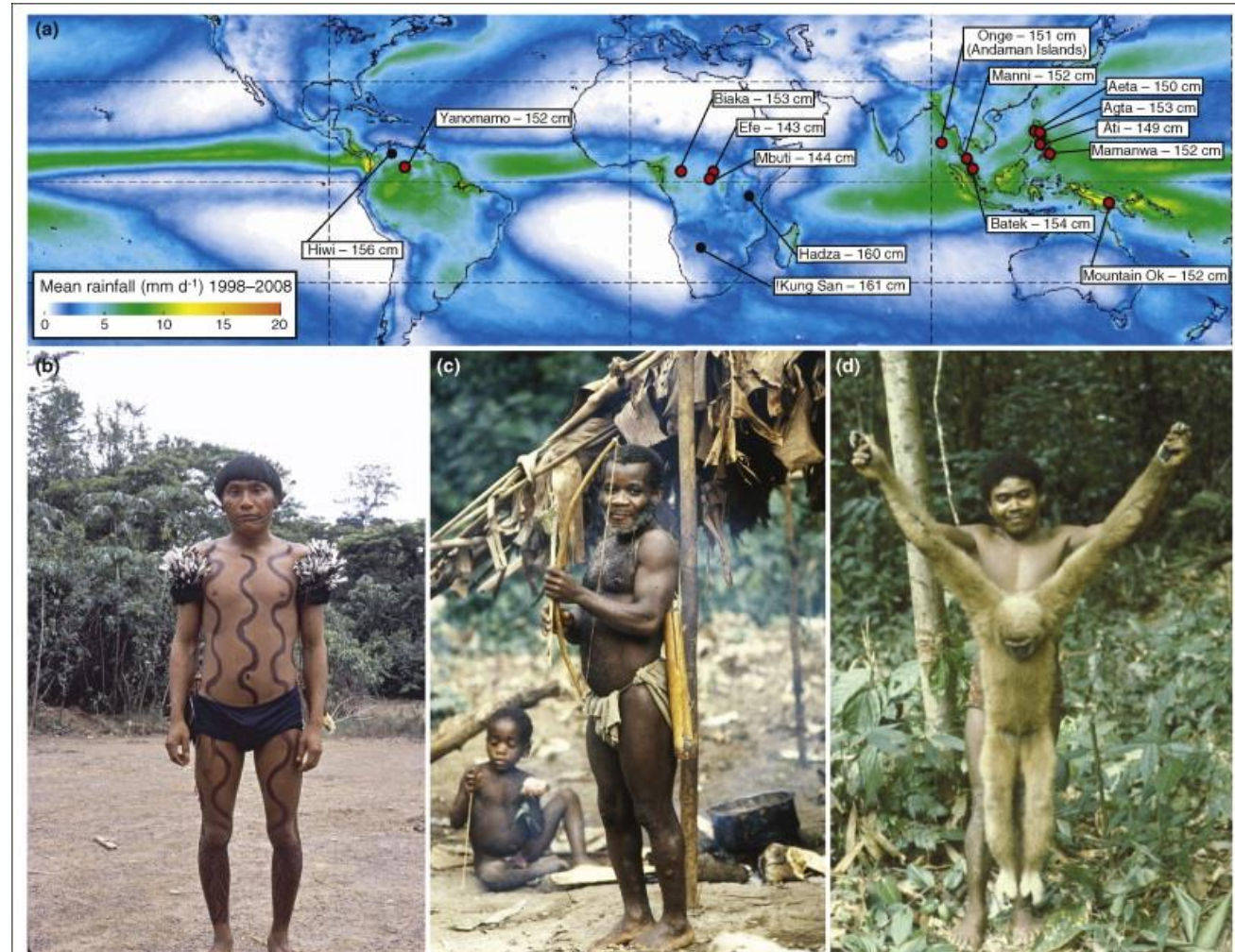


1. How does it happen? | 2. Selective pressures | 3. Adaptive variation

Rainforests:

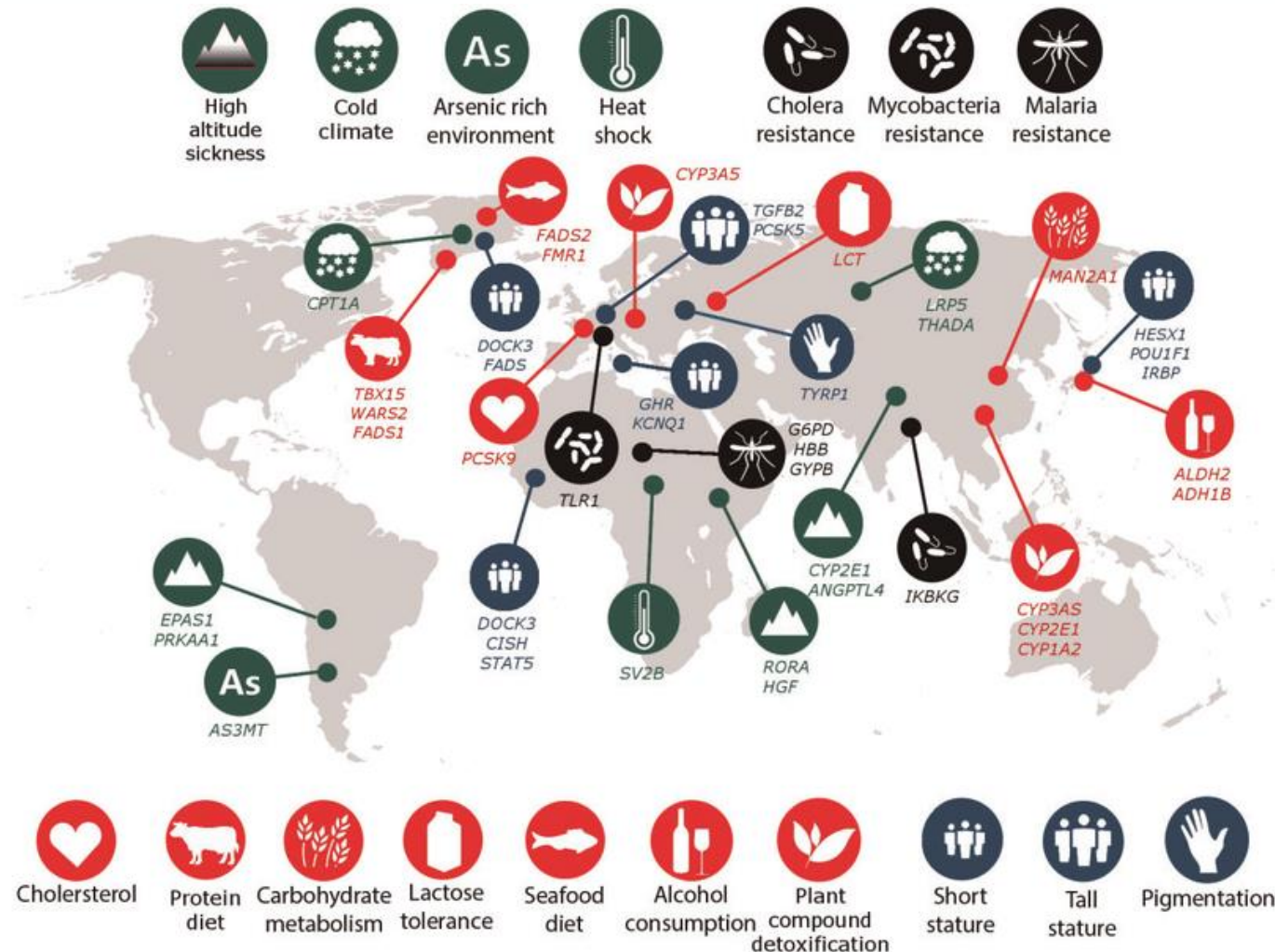
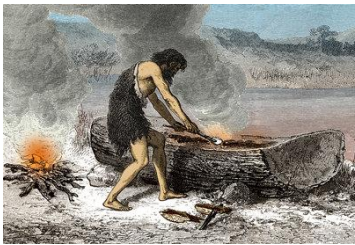
- High temperature *and* humidity
- Parasites and other pathogens
- 'Pygmy' phenotype: average height of <150cm in adult males
 - Convergent evolution
 - Limited food; heat stress; immune response; **tradeoff between growth/maintenance and reproduction**
 - Strong selection on small number of loci
 - Chromosome 3:
 - DOCK3, CISH
 - HESX1 (development of anterior pituitary; growth hormone). In Central African, but not Ugandan, RFHG.

Tradeoff
in
ontology



TRENDS in Ecology & Evolution

Culture as a driver of local adaptation



Elhaik et al. (2017),
based on Fan et al.
(2016)

The key question: in such a genetically homogenous but globally distributed species, does **local adaptation** effectively only happen through **short-term processes** (culture, homeostatic responses, and acclimatization)?

Natural
selection
in genome
due
to culture

Two main takeaways

1. The human genome is (uniquely?) **buffered** from intense selection. Most human genetic variation is selectively neutral. Culture plays an important role, but there are also arguments that humans are exceptionally developmentally adaptable.
2. However... there are stressors that surpass short-term processes! One of the most important stressors (measured in the number of known genetic adaptations *and* the strength of selection) is **disease**.